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Ship–shore drill-based training for GNSS spoofing response

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ABSTRACT

Advancing technology onboard ships brings with it concerns regarding cyber threats and vulnerabilities. A comprehensive cyber security approach is required to prevent such cyber risks. By performing drills, the effectiveness of these measures can be evaluated. Moreover, the cyber awareness of employees can be improved, and their competence in responding to cyber incidents can be assessed. The essential objective of this study is to design effective training for crew members and office personnel at ship operators to address Global Navigation Satellite Systems (GNSS) spoofing risks onboard ships and to observe their responses to such incidents during the drills. To this end, the Maritime Cyber Security (MarCy) training programme, which was developed based on the Critical Events Model (CEM), was employed to deliver comprehensive training to the employees of two tanker operators. These training sessions particularly focused on a ship–shore combined drill involving 10 different third-party stakeholders. Following the drills, a lecture covering GNSS-related cyber risks and the cyber security procedures of the tanker operators was delivered. The findings of this study revealed that the two participating ship operators were not adequately prepared to prevent cyber risks onboard ships. Moreover, the participating operators encountered difficulties in obtaining sufficient support from stakeholders during the simulated GNSS spoofing incidents. This study highlights the urgent need for maritime authorities to introduce additional regulations to improve preparedness against emerging cyber threats.

1. Introduction

Given the increasing digitalisation of systems onboard ships, cyber risks have become a growing concern for the maritime industry. To effectively address these risks, it is essential to consider the three fundamental elements of cyber security, including people, process, and technology. Even if technological solutions and robust procedures are developed, cyber security awareness of the human element is still vital to prevent cyber incidents. The human element could be improved through effective instruction strategies, such as lectures, discussions, and demonstrations.

Another important instruction strategy is drills. A drill offers opportunities to test systems and train personnel through the simulation of realistic incidents. The International Maritime Organisation (IMO) defines a drill as “a form of exercise which provides the opportunity to practice elements of the system. Drills are carried out under realistic conditions while allowing for instruction and training.” (IMO, 2014) IMO also defines exercise as “a test of the emergency response arrangements under as near realistic conditions.” (IMO, 2014)

A collaboration is typically required to respond to marine incidents effectively. Therefore, to increase the effectiveness and realism of drills,

it is essential to integrate both the ship’s crew and shore-based stakeholders. By simulating this coordination, drills can help identify deficiencies in company procedures, the effectiveness of communication protocols, and the adequacy of stakeholder support. This combined drill approach is typically called in the maritime industry *ship-to-shore drill* (Coach, 2021), *ship-shore drill* (SQE, 2026), or *ship-shore combined drill* (Denizcilik, 2026).

1.1. Objective, methodology, and contributions

The essential objective of this study is to propose a comprehensive and effective training programme for crew members and office staff at ship operators to address Global Navigation Satellite Systems (GNSS) spoofing risks onboard ships and to observe their responses to such incidents. This study also has two sub-objectives, as follows:

- to evaluate and improve the effectiveness of the Maritime Cyber Security (MarCy) training programme;
- to examine the documentation-based preparedness of ship operators against cyber risks.

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For these purposes, all materials regarding cyber security in the Safety Management System (SMS), Ship Security Plan (SSP), and Ship Security Assessment (SSA) of two tanker operators were investigated. Subsequently, the MarCy training programme (Oruc et al., 2024a), which was developed with the Critical Events Model (CEM) approach (Nadler and Nadler, 1982), was employed to perform a training session for each operator. In the first part of the training, a ship-shore combined drill for the GNSS spoofing scenario was conducted with the attendance of maritime stakeholders. In the second part of each training session, the cyber risks of GNSS and the cyber security procedures of tanker operators were delivered through a comprehensive lecture. The effectiveness of the training programme was assessed using pre- and post-training knowledge assessments, allowing the impact of the training on learners' knowledge to be quantitatively evaluated.

The effectiveness of the MarCy training programme has been evaluated with real-world results before (Oruc et al., 2024b). The lecture, discussion, and case study instruction strategies were employed in the previous study. However, the MarCy training programme also suggests the implementation of the drill approach for maritime cyber security training. Through performing these two ship-shore combined drills, the effectiveness of the drill method was also evaluated. Moreover, all steps of the MarCy training programme were evaluated again.

To the best of our knowledge, this is the first study in the literature regarding a cyber security drill on ships in service. The study revealed that the two participating ship operators were unable to provide a rapid and effective response to the simulated GNSS spoofing scenarios, however, they were able to continue their operations using traditional navigation methods. It also demonstrated that the cyber security procedures examined at the two participating ship operators had deficiencies, and their stakeholders couldn't provide support during a potential cyber incident. By considering the findings, observations, and suggestions in this study, maritime stakeholders can effectively perform cyber security training sessions, particularly for the GNSS spoofing drill, onboard ships. Moreover, they may improve their cyber security procedures.

1.2. Scope

This study is limited to the evaluation of GNSS spoofing preparedness based on two distinct training sessions conducted. The findings, observations, and insights are specific to the conditions and context of these training sessions. The drills simulated the operational consequences of a GNSS spoofing incident and did not involve the technical generation or injection of spoofed GNSS signals into onboard systems. The study focuses on the preparedness and responses of shipboard crew and shore-based personnel. Results could vary with the involvement of different actors, such as alternative stakeholders and personnel groups. Therefore, the results presented in the study should not be generalised across the entire maritime industry. Moreover, different cyber incident scenarios may lead to different results. Finally, this study does not include a systematic gap analysis of the cyber security procedures implemented by ship operators.

1.3. Structure

This article is organised as follows. Section 2 provides background information on the cyber risks of GNSS and the importance of drills in the maritime sector. Related works in the literature are reviewed in Section 3. Section 4 describes the methodology employed in detail. Section 5 assesses the cyber security-related procedures of ship operators. The findings and observations from the performed training are presented in Section 6. Section 7 discusses the findings and implications of the study. Finally, Section 8 summarises the research and outlines potential directions for future work.

2. Background

The Background section provides a foundation for this study by presenting technological, operational, and regulatory aspects. The role of GNSS for ships and its cyber security risks are briefly explained with experienced incidents. Furthermore, the importance of drills is elaborated in the maritime industry. The integration of these perspectives establishes the technical and procedural context necessary to understand the motivations and scope of this study.

2.1. GNSS and its vulnerabilities

GNSS is used in a wide range of applications to provide continuous and accurate data. Four GNSS variants are operated worldwide, including the Global Positioning System (GPS) operated by the United States (U.S.), the Globalnaya Navigatsionnaya Sputnikovaya Sistema (GLONASS) operated by Russia, the BeiDou Navigation Satellite System (BDS) operated by China, and the European Union's Galileo (Jan and Tao, 2016). The performance of GNSS can be enhanced through satellite-based (i.e. Satellite-based Augmentation System (SBAS)) or terrestrial-based (i.e. Differential GNSS (D-GNSS)) systems (Ibanez et al., 2018).

A GNSS receiver provides continuous information for ships, including Course over the Ground (COG), Speed over the Ground (SOG), latitude/longitude position, and Universal Time Co-ordinated (UTC) (IMO, 2000). Various ship components, such as the Automatic Identification System (AIS), Electronic Chart Display and Information System (ECDIS), gyro compass, and Radio Detection And Ranging (RADAR), depend on the data transmitted from the GNSS receiver (Oruc et al., 2022). Therefore, it is a critical component in the safe navigation of ships.

At sea level, GNSS signal strength for civilian applications is approximately -160 dBW (Deng et al., 2009). Since the signal is too weak, it is inherently vulnerable to two main cyber attacks, namely GNSS spoofing and GNSS jamming. A spoofing attack causes a GNSS receiver to show false information (Filic, 2018). A jamming attack prevents a GNSS receiver from receiving signals (Świerczyński et al., 2016). The impact of both attacks depends on the sensitivity of the GNSS receiver, the power of the attacker's device, and the distance between the ship and the malicious device (Glomsvoll and Bonenberg, 2017). As a GNSS spoofing attack is more difficult to detect by seafarers, it is typically considered more dangerous (Liu et al., 2018).

GNSS attacks have been on the agenda of the maritime sector for many years. In 2016, a GPS jamming attack affected approximately 280 vessels sailing off the coast of South Korea and caused them to return to ports (Graham, 2026). More than 20 vessels in the Black Sea were subjected to a GPS jamming attack in June 2017 (Goward, 2026). A report published in 2019 unveiled that a total of 1311 civilian ships were targeted by GNSS spoofing attacks between 2016 and 2018 (C4ADS, 2019). On June 16, 2025, two oil tankers collided in the Gulf of Oman. Although it was officially attributed to a navigation error, it may have been influenced by GPS jamming or spoofing (Cydom, 2026). In July 2025, researchers identified near-daily GPS jamming and spoofing across the Baltic Sea (Höller, 2026). In August 2025, a GNSS spoofing attack affected nearly 98 vessels in Russia's Far East (Diakun, 2026). GNSS attacks have now become a common occurrence in the maritime sector. As of 2026, GNSS jamming and spoofing have become persistent and widespread threats in the maritime domain rather than isolated incidents. Today, the intensity of these attacks can even be monitored on daily updated global maps (GPSwise, 2026; Wiseman, 2026) as seen in Fig. 1, in addition to the incidents reported in the news.

2.2. Drills and cyber security drills in the maritime sector

Drills in the maritime sector are divided into two groups: mandatory and voluntary. Under IMO regulations, various mandatory drills are

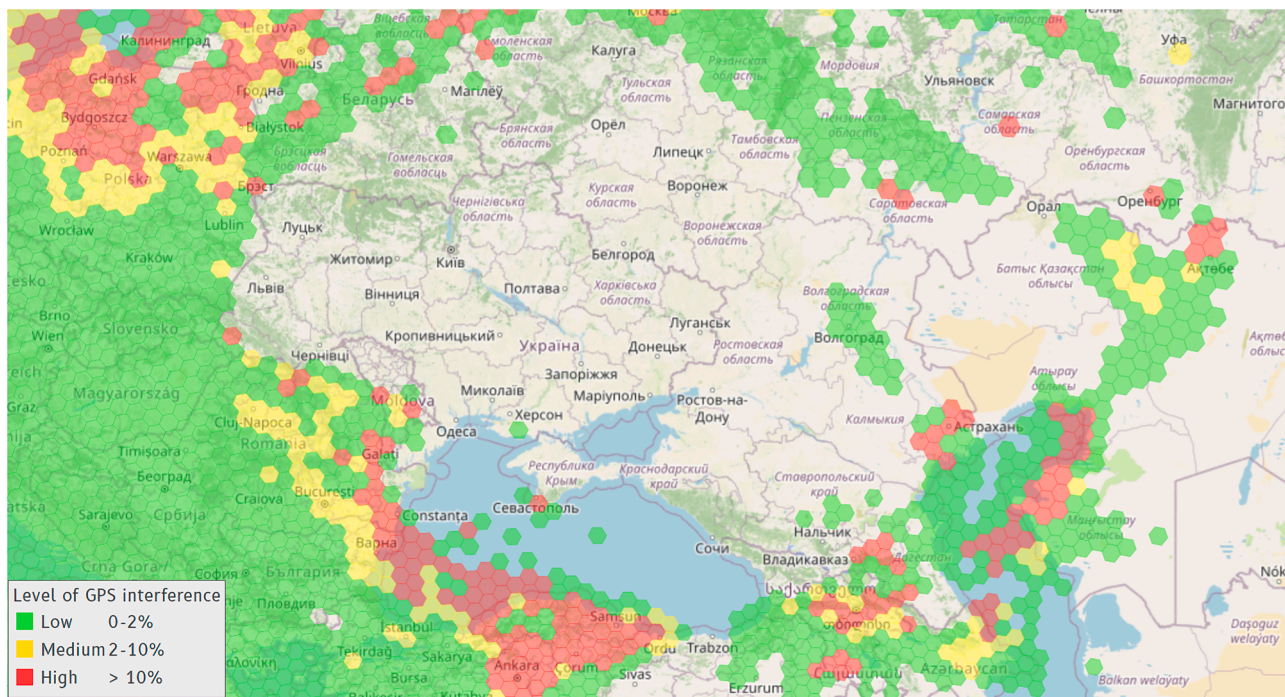


Fig. 1. Example of a GNSS interference map (on 05 July 2026) (Wiseman, 2026).

available, such as fire drills, abandon ship drills, and enclosed space entry drills (IMO, 2017a). The IMO also defines how such drills must be performed and at what frequencies (IMO, 2017a). However, recognised maritime organisations, such as flag states, classification societies, and associations, are allowed to improve these requirements. Implementation of these drills is verified by relevant parties. Ship operators are also allowed to conduct additional drills based on their own needs. Moreover, they can improve the criteria set by the IMO or other relevant organisations.

Drills are not always performed onboard ships solely by the ship's crew. Under the International Ship and Port Facility Security (ISPS) Code, a ship-shore combined drill must be conducted at least once each calendar year, with no >18 months between drills, with the attendance of third parties such as relevant authorities of contracting governments and port facility security officers (IMO, 2002a, 2002b).

Vetting programmes, including the Ship Inspection Report Programme (SIRE) 2.0, the Chemical Distribution Institute (CDI), the Tanker Management Self Assessment (TMSA), and RightShip, play a key role in the business life of ship operators. According to a TMSA requirement (i.e. Rule 11.4.2), a ship-shore combined drill must be conducted by tanker operators (OCIMF, 2017). Moreover, SIRE 2.0 (i.e. Rule 5.1.1) requires ship operators to develop shipboard emergency plans that include all emergency scenarios and integrate ship and shore response (OCIMF, 2022).

Under IMO regulations, cyber security drills are not mandatory. However, the IMO recommends that ship operators assess cyber security knowledge of seafarers employed through cyber security drills (IMO, 2025). The IMO also recommends various guides and standards to support ship operators in their cyber risk management. One of these guides, specifically published for the needs of the maritime sector, recommends conducting cyber security drills for seafarers (BIMCO, 2024). According to this guide, masters and senior officers in particular should be reinforced through drills to effectively respond to cyber incidents onboard ships.

Another standard suggested by the IMO is UR (stands for Unified Requirement) E26 (IACS, 2023) published by the International Association of Classification Societies (IACS). This standard requires ship operators to conduct cyber security drills for both seafarers and

shore-based personnel. Because of IACS requirements, IACS member classification societies, such as Bureau Veritas (BV) (BV, 2024), ClassNK (stands for Nippon Kaiji Kyokai) (ClassNK, 2024), and Det Norske Veritas (DNV) (DNV, 2024), have also reflected these drill requirements in their rule booklets.

Compliance with UR E26 is mandatory for ships contracted for construction on or after 1 July 2024. Given that over 90% of the world's cargo-carrying tonnage by ships is covered by IACS member classification societies (IACS, 2026), it can be expected that cyber security drills would be mandatory for almost all newbuildings worldwide.

According to a U.S. Coast Guard regulation (CFR Part 101 Subpart F Cybersecurity, 2025), cyber security drills must be conducted at least twice a year on U.S.-flagged vessels certified under the ISPS Code. These drills are designed to evaluate the ship crew's competence in effectively implementing cyber security plans and their preparedness against potential cyber threats. Individual response procedures, communication, and coordination protocols are assessed during the drills. All identified deficiencies are documented and addressed.

Vetting programmes include cyber security training requirements for crew or for both crew and shore-based staff (i.e. Rule 13.2.4 in TMSA (OCIMF, 2017), 12.12 in CDI (CDI, 2023), 7.5.1 in SIRE 2.0 (OCIMF, 2022), and 12.7 in RightShip Inspection Ship Questionnaire (RISQ) (RightShip, 2025)). To ensure compliance with these requirements, many ship operators incorporate cyber security drills into their SMS.

3. Related work

The Related Work section discusses studies aimed at creating realistic environments for cyber security training for maritime stakeholders. Reviewing these studies provides an understanding of current gaps and findings in the literature. Existing studies typically suggest the use of marine simulators and cyber ranges to improve the cyber readiness of the human element. However, the literature lacks findings and observations on cyber security drills onboard ships in service with the participation of various maritime stakeholders.

Lovell and Heering (2019) conducted Exercise Neptune at the Tallinn University of Technology. The objective of this study was to improve the situational awareness and readiness of ship crews against cyber risks

onboard ships. In the exercise, a full mission bridge simulator was employed, and various Open Source Intelligence (OSINT) tasks were carried out. Therefore, technical vulnerabilities and the human factor were integrated into the exercise. The exercise unveiled that AIS, ECDIS, and Global Maritime Distress and Safety System (GMDSS) could be compromised by malicious actors. Moreover, 7536 usernames and passwords used by the personnel of North Atlantic Treaty Organization (NATO) warships were successfully captured. The exercise also highlighted the necessity of cyber security drills and unified ship-shore security policies.

Erstad et al. (2023) evaluated the effectiveness of marine simulators in improving cyber security awareness. To this end, the Human-Centred Design (HCD) methodology was employed. By considering learners' expectations, the authors designed a cyber attack scenario targeting ballast water management systems. Using a marine simulator, the response, communication, and coordination capabilities of a ship's crew and the office staff of a ship operator were thoroughly evaluated. The findings of the study revealed that simulator-based training grounded in HCD improves the situational awareness and coordination capabilities of participants in cyber security incidents.

Potamos et al. (2023) proposed an innovative training curriculum for the maritime sector to enhance cyber readiness against ransomware attacks. To this end, the Maritime Cyber Range (MARTE) was utilised. In the study, the authors integrated lectures, tabletop exercises, demonstrations, and hands-on cyber range drills simulating realistic ransomware scenarios, such as attacks on ECDIS and ship-to-shore networks. The findings revealed that a unified curriculum covering both Information Technology (IT) and Operational Technology (OT) systems improves the situational awareness, decision-making, and incident response coordination of relevant stakeholders.

La Vallée et al. (2022) proposed a federated maritime cyber training framework. The objective of the study was to demonstrate interconnected virtual environments for creating realistic and complex ship-shore cyber security training scenarios. To this end, the authors developed a hands-on exercise simulating an attack on ECDIS systems onboard a passenger ship. The multi-network environment replicated the ship's navigation, service, and passenger networks. The findings revealed that federated scenarios can significantly improve the situational awareness and defensive decision-making of learners. The study also demonstrated that federated cyber range training can be extended to ship-shore response drills.

Tam et al. (2020) discussed the effectiveness of cyber ranges in maritime cyber security applications. To this end, the authors reviewed existing literature from other sectors, such as defence, where cyber ranges are actively used, and examined potential implementation challenges for the maritime industry. Although the study does not present any experimental results, the authors argue that cyber ranges can be utilised for research and educational purposes in maritime cyber security. It is particularly highlighted that it could be beneficial to evaluate the effectiveness of collaboration between ship crews and shore-based staff under realistic cyber attack scenarios.

Svilicic et al. (2020) conducted a cyber risk assessment on two chemical tankers in service. To this end, the authors adopted a mixed-method approach combining crew interviews and vulnerability scanning of RADARs. The scanning tools detected various vulnerabilities, particularly in the operating systems underlying the RADAR software. The study also revealed that cyber security policies and procedures in the SMS were insufficient. Moreover, crew members were not familiar with system-specific cyber risks or effective incident response procedures.

The literature review demonstrates that it is required to enhance the human element to effectively prevent cyber risks in the maritime sector. As previously discussed in the papers, the authors strove to create realistic environments to improve the cyber awareness and response capabilities of crew members and shore-based staff. To this end, they employed marine simulators. Moreover, some studies have been

conducted onboard ships to evaluate the actual preparedness of ship operators in addressing cyber risks. However, to the best of our knowledge, there are no studies on cyber security drills conducted onboard ships in service. In addition, experimental results on cyber security training for maritime personnel remain limited.

In this study, we present findings and observations on cyber security drills based on GNSS spoofing. The scenario was designed as a ship-shore combined drill involving two ship operators and additionally 10 different stakeholders. Therefore, in the event of GNSS spoofing, the actual behaviours of not only ship crew and shore-based staff but also service providers, classification societies, maritime authorities, ship agencies, and marine insurers could be evaluated. Furthermore, we presented our observations regarding cyber security policies and procedures of two tanker operators. This constitutes an additional contribution that distinguishes our study from the existing literature.

4. Methodology

This section explains the methods employed in this study. The MarCy training programme was designed to address the cyber security training needs of maritime stakeholders. Its effectiveness has been demonstrated both through expert evaluation (Oruc et al., 2024a) and actual field results (Oruc et al., 2024b).

This study was built upon the MarCy training programme. Two distinct training sessions were designed and conducted using the MarCy. However, the MarCy alone was not sufficient to address the research objectives outlined in Section 1.1. Therefore, a process integrating the steps of the MarCy was designed and implemented, as described in Section 4.2. In addition, the Thematic Data Analysis was employed during the data analysis process.

In this study, the term *operator* refers to a ship operator with whom the training sessions were conducted collaboratively to address the research objectives outlined in Section 1.1. The term *learner* refers to a participant who attended the training session. The term *lecturer* refers to the author of this study, who delivers lectures to the learners.

4.1. MarCy training programme

In the literature, various models have been proposed for designing and developing training programmes (Aris et al., 2017; Bobbitt, 2009; Mirriahi et al., 2015; Schagaev et al., 2010). The Process of Training approach was initially proposed in 1965 (Nadler and Nadler, 1982). In 1982, this model was finalised and introduced as the Critical Events Model (CEM) (Nadler and Nadler, 1982). Today, this approach is also known as Nadler's Model (Erasmus and van Dyk, 1999). The CEM can be applied to the design of training programmes, ranging from formal university courses to the training needs of various organisations.

The MarCy training programme was developed by applying the Critical Events Model (CEM) to specifically address cyber security needs in the maritime industry (Oruc et al., 2024a). During its implementation, the CEM was slightly adapted to meet the specific requirements of maritime cyber security training design. One key difference is that MarCy adopts a modular approach. Modular training addresses the specific training needs of learners and, as it minimises disruption to business operations, represents a cost-effective approach (Singh et al., 2012). Therefore, it is particularly preferred for vocational training needs (French, 2015). Moreover, it enables the design of role-specific training curricula. This role-specific training approach is recommended by recognised maritime organisations to address cyber security training needs in the maritime sector (OCIMF, 2022; BIMCO, 2024; Witherbys, 2025; DSCA, 2020). The MarCy training programme comprises 12 training modules, as presented in Table 1.

In designing a training session, lecturers can select one or more modules for learners by considering their specific training needs and job responsibilities. Moreover, if necessary, lecturers can develop additional training modules by following the phases of the MarCy training

Table 1
Training modules with their objectives (Oruc et al., 2024b).

Code	Title	Objective
M1	Basic cyber security	The attendees will be familiar with the basics of cyber security.
M2	Advanced cyber security	The attendees will have advanced knowledge of cyber security.
M3	Regulatory requirements	The attendees will learn regulations for cyber security.
M4	Vetting requirements	The attendees will learn cyber security requirements in vetting programmes.
M5	Critical deck systems	The attendees will learn cyber risks in the critical deck systems.
M6	Critical engine systems	The attendees will learn cyber risks in the critical engine systems.
M7	Other critical systems	The attendees will learn cyber risks in other critical systems.
M8	Cyber security investments	The attendees will be able to decide cyber security investments.
M9	Cyber security practices	The attendees will have practical experience in implementation.
M10	Cyber security management	The attendees will be able to manage cyber security issues in a company.
M11	Advanced skills	The attendees will have advanced technical skills.
M12	Autonomous ships	The attendees will learn the potential cyber risks of autonomous ships.

programme presented in Table 2.

During the development of the MarCy training programme, it was evaluated twice using the Delphi method (Dalkey and Helmer, 1963). A total of 19 scientists and professionals with expertise in maritime operations, maritime training, and maritime cyber security participated in the evaluation process. Based on the comments and recommendations received, the MarCy training programme was further improved and subsequently published as a scientific paper.

The first article (Oruc et al., 2024a) on the MarCy training programme was based primarily on expert opinions and lacked real-world results. Subsequently, four training sessions involving different learner groups (i.e. maritime students, non-maritime students, employees of a classification society, and representatives of various shipping companies) were conducted to evaluate the effectiveness of each stage of the MarCy training programme. The findings revealed that MarCy was a highly effective programme for maritime cyber security training. Nevertheless, in the second article (Oruc et al., 2024b), we proposed various recommendations to further improve the programme based on the experience gained from the training sessions conducted.

4.2. Implementation process of the MarCy training programme

In this study, we developed a process by integrating the steps of the

Table 2
Phases of the MarCy training programme (Oruc et al., 2024a).

Phase	Function
Step 1: identify the needs of the organization	Modules are identified by considering the needs of the organization.
Step 2: specify job performance	Roles and responsibilities of learners are investigated.
Step 3: identify learner needs	Modules are mapped with roles by considering responsibilities.
Step 4: determine objectives	Objectives and the learning outcomes of the modules are identified.
Step 5: build curriculum	A curriculum is created for the modules.
Step 6: select instructional strategies	Instruction modalities are identified.
Step 7: obtain instructional resources	Training resources required are analysed.
Step 8: conduct training	It is identified how to perform the training.
Step 9: evaluation and feedback	The effectiveness of the designed training is verified.

MarCy to achieve the research objectives. This process also enabled us to analyse the findings and observations both quantitatively and qualitatively. The steps of the process are presented below and illustrated in Fig. 2.

1. identification of ship operators;
2. conducting a planning meeting;
3. analysis of training expectations;
4. designing the training;
5. conducting the training;
6. evaluation of the findings;
7. reporting the findings and improving the MarCy.

Fig. 2 also illustrates the integrated steps of the MarCy training programme presented in Table 2. The step numbers shown in the figure (e.g. Step 1–2–3, Step 4, and Step 5–6–7) correspond to the relevant steps of the MarCy training programme and demonstrate how they were integrated throughout the implementation process.

4.2.1. Identification of ship operators

To conduct training sessions addressing research objectives, it was necessary to identify ship operators willing to collaborate in this study. To this end, invitation letters explaining the purpose of the study and the expectations from ship operators were sent via e-mail to multiple ship operators. Two tanker operators accepted the invitation. In the remainder of this article, they are referred to as *Operator A* and *Operator B*.

4.2.2. Conducting a planning meeting

This stage incorporated Step 1 (i.e. identify the needs of the organisation), Step 2 (i.e. specify job performance), and Step 3 (i.e. identify learner needs) of the MarCy training programme. Planning meetings were held separately with each operator. In-person meetings were conducted with the responsible shore-based staff of the ship operators to organise the training sessions. During this stage, the objectives of the training, expectations from the ship operators, and the potential benefits for the operators were explained by the authors.

Subsequently, operator details, the details about potential learners (i.e. ship crew and shore-based staff), cyber security-related documents, operators' training expectations, technical details of the ships on which the training would be conducted (e.g. GNSS-dependent components), and the planned training dates were discussed. Moreover, the authors presented their recommendations for the training process. During this stage, all cyber security-related materials within the SMSs of the operators were provided to the authors, including policies, procedures, checklists, risk assessments, and posters.

4.2.3. Analysis of training expectations

This stage incorporated Step 4 (i.e. determine objectives) of the MarCy training programme. During this process, the training requirements of the ship crew and shore-based staff were identified. Based on these findings, the learning objectives and expected outcomes of the training were determined.

At this stage, all materials received from the ship operators were also analysed. In addition, the procedures to be followed by the ship's crew and shore-based staff in the event of a potential GNSS spoofing attack against the ship were identified and reviewed.

4.2.4. Designing the training

This stage incorporated Step 5 (i.e. build curriculum), Step 6 (i.e. select instructional strategies), and Step 7 (i.e. obtain instructional resources) of the MarCy training programme. The drill and lecture components were determined as the instructional strategies at the beginning of the study.

For the drill scenario, a situation was designed in which the GNSS receiver displayed an incorrect position. However, the anomaly was not

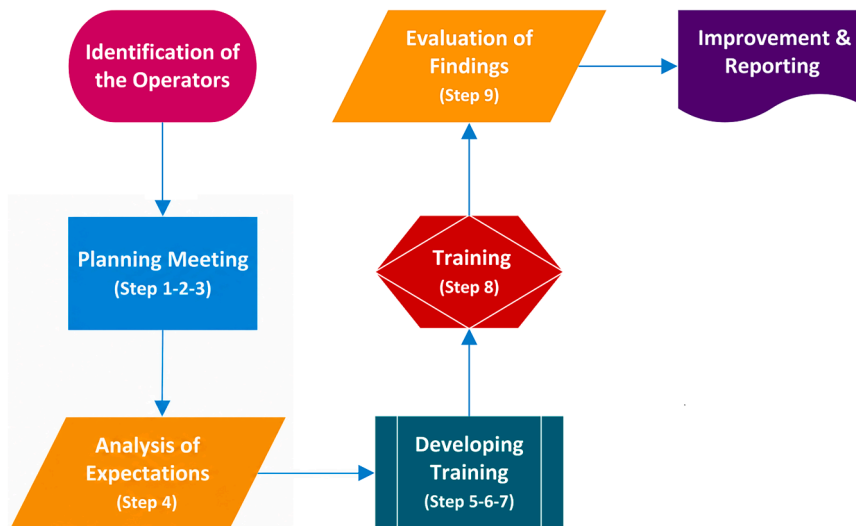


Fig. 2. Illustration of the methodology.

directly identified as a cyber attack, as it could also have resulted from a GNSS receiver failure. The learners in the drill should have decided whether it was a cyber attack or a failure. The GNSS spoofing incident was simulated at the operational level. No spoofed GNSS signals were generated or injected into the vessel's onboard GNSS systems.

By considering the relevant procedures provided by the operators and the technical details of the ship, a presentation for the lecture component of the training was prepared. This enabled the delivery of tailored training to the ship's crew and shore-based staff.

Moreover, an assessment survey was developed at this stage. The survey consisted of five sections: learners' background, quiz, test, operator evaluation, and feedback on the training. In this study, the term *Quiz* refers to an examination conducted before the training, whereas *Test* refers to an examination conducted after the training.

At this stage, the requirements necessary for organising the training were also communicated to the operators. These included arranging a meeting room and testing the equipment (e.g. cameras and microphones) and internet connections on both the ship and office sides to ensure a stable and uninterrupted MS Teams connection.

4.2.5. Conducting the training

This stage incorporated Step 8 (i.e. conduct training) of the MarCy training programme. The drill was conducted with the participation of the ship crew, shore-based staff, and third-party stakeholders. Following the drill, a tailored lecture was delivered to the ship crew and shore-based staff of operators. The lecture was delivered to the shore-based staff on-site and simultaneously to the ship crew via MS Teams. During the training sessions, the entire process was recorded through MS Teams.

Through the assessment survey, information regarding the learners' backgrounds was collected, and the quiz section was completed by the learners before the training. After the training, the remaining sections of the assessment survey were completed by the learners.

4.2.6. Evaluation of the findings

This stage incorporated Step 9 (i.e. evaluation and feedback) of the MarCy training programme. During this process, the effectiveness of the designed training was evaluated. The qualitative data collected from the assessment surveys and training video recordings (transcripts) were analysed using Thematic Data Analysis. The open-source software QualCoder (QualCoder, 2026) was employed to facilitate the effective implementation of the method.

Thematic Data Analysis is a widely used method for identifying, analysing, and reporting patterns (themes) within qualitative data

(Braun and Clarke, 2006). This approach facilitates the analysis of complex datasets. In this study, the six-phase framework proposed by Braun and Clarke (2006) was adopted:

1. familiarising yourself with your data;
2. generating initial codes;
3. searching for themes;
4. reviewing themes;
5. defining and naming themes;
6. producing the report.

4.2.7. Reporting the findings and improving the MarCy

The preparedness and responses of the participating maritime stakeholders to the simulated GNSS spoofing incidents are presented in this study. Moreover, various recommendations are provided to further improve the MarCy training programme based on the findings and observations obtained.

The clarity and academic readability of the paper were enhanced through the use of artificial intelligence (AI) writing assistants called ProWritingAid (ProWritingAid, 2026) and Grammarly Pro (Grammarly, 2026). Both writing instruments aided in the refinement of the manuscript by offering recommendations on grammar, writing style, sentence structure, and overall organisation.

4.3. Ethical considerations

Before the training sessions, all participants were verbally informed by the researcher about the purpose and nature of the study, the use of the assessment survey data for scientific research and publication, and the recording of the training sessions. The assessment survey also included a written statement informing participants that their responses would be used anonymously for scientific research and publication. Although participation in drills is part of the participants' regular professional duties, participants were informed that their participation in the study-specific drill and assessment survey was voluntary. The representatives of the ship operators also explicitly emphasised that participants could decline to participate without any consequences. No participant declined to participate.

No personal or demographic information, such as age, gender, nationality, or religion, was collected, and no signatures were requested. Participants were informed that the training sessions would be recorded through MS Teams. Access to both the assessment surveys and training recordings was limited to the researchers. The assessment surveys and training recordings will be deleted following publication of the study.

They will not be uploaded to any public repository or made publicly available.

5. Assessment of the procedures of the ship operators

This section investigates the cyber security-related procedures of the ship operators. It does not provide a comprehensive gap analysis. In other words, neither the compliance of the procedures with current regulations and standards nor their practical effectiveness is deeply evaluated. Instead, this section reflects the documented actions of the operators at the documentation level.

To this end, cyber security-related materials within the SMSs of both operators were requested. Moreover, the SSP and the SSA are confidential documents. Therefore, the operators were asked whether any cyber security-related procedures were included in the SSP and SSA. All collected documents were analysed and reported in this study.

Both operators share several similar practices and perspectives regarding cyber security. For instance, a hard-copy cyber security plan is available onboard ships for both operators. Neither operator receives consultancy services related to OT security onboard ships. Both operators provide online cyber security training for shore-based staff and ship crew through a maritime training company accredited by an IACS member classification society. The training provided is typically IT-oriented. Cyber risks of OT systems onboard ships are not discussed. Moreover, such training courses are not operator-specific. At the end of the training, those who pass the exam are certified. On the other hand, those who fail repeat the training until they succeed.

According to the U.S. Coast Guard, a Cyber Security Officer (CySO) is responsible for the implementation, maintenance, and auditing of the cyber security plan and cyber incident response plan (Cybersecurity in the Marine Transportation System, 2025). In this study, the CySO role is categorised into two groups: the Company Cyber Security Officer (CCySO) and the Ship Cyber Security Officer (SCySO). The SCySO refers to the individual responsible for cyber security matters onboard ships, whereas the CCySO refers to shore-based personnel responsible for overseeing cyber security management within a ship operator. The operators involved in this study may use different titles for these roles. However, the above designations are used consistently throughout this study to avoid conceptual ambiguity.

In both companies, the IT Manager assumes the role of CCySO, as IT managers are generally more familiar with IT systems and equipment than other shore-based staff. They are responsible for the IT security of both the ships and the operator's offices. Similarly, both operators assign the role of SCySO onboard ships to 2nd Officers. Given that 2nd Officers are generally younger and more familiar with using computers, they are considered more suitable for this responsibility. Operator B also argues that, since 2nd Officers are responsible for the navigation of the ship, they might detect cyber security risks of navigation equipment on the bridge compared to other crew members.

Although both operators have designated the 2nd Officer as the SCySO, no recommendations supporting the appointment of 2nd Officers to this role were identified in recognised guidelines. On the other hand, several guidelines suggest that the master (Witherbys, 2025), chief officer (Witherbys, 2025), chief engineer (BIMCO, 2024), electro-technical officer (Witherbys, 2025), or Ship Security Officer (SSO) (BIMCO, 2024) may be assigned to this position.

Over the years, both operators have experienced personnel changes on the shore side. As a result, the current responsible personnel are not fully aware of the basis upon which the procedures were originally developed. Consequently, newly assigned personnel generally revise the procedures without a comprehensive understanding of the rationale behind them. This situation also applies to the roles of SCySO and CCySO. Operator A stated that these assignments were made based on the recommendations of a maritime consultancy company. Operator B acknowledged that the appointments currently align with SIRE 2.0 requirements but noted that these roles had existed even before the

implementation of SIRE 2.0, without any clear understanding of why they had originally been established. Moreover, neither operator provides additional cyber security training to CCySOs and SCySOs beyond the training given to other officers onboard ships or shore-based staff. Although the responsibilities of the SCySO are documented within the SMS, their effectiveness has not been verified in operational practice.

In this study, learners were also asked to provide their opinions regarding the cyber security practices of the operators for whom they work. The questions asked and the learners' responses are presented in Table 3. This information is important, as it provides insights into how cyber security practices are perceived by personnel directly involved in daily operations.

Percentages were calculated based on the valid responses to each question, as some learners skipped certain questions. Due to rounding to whole numbers, the percentages for some questions may sum to slightly above or below 100%.

Although learners generally consider the generic training, they received to be sufficient, their responses indicate a desire for more specific training on cyber security policies and procedures. In addition, they prefer ship-shore combined drills to be repeated using different attack scenarios. They also believe that the ship operators for whom they work should allocate more resources to address cyber security needs.

The following subsections present operator-specific cyber security practices and procedures. The cyber security procedures of both operators, including policies, plans, training programmes, and risk assessments, are explained. Moreover, various observations regarding the implementation of these procedures are reported.

5.1. Operator A

The cyber security policy of Operator A applies to all office staff and crew members onboard ships. It focuses on preventive measures for all IT and OT assets against internal and external cyber threats. The policy also emphasises regulatory compliance, risk management, data backup systems, contingency planning, personnel training, and incident reporting.

The cyber security plan was purchased from a maritime consultancy company and includes comprehensive procedures to protect IT and OT systems onboard ships. In accordance with the plan, the top management of the operator is responsible for implementing all security measures in line with international requirements, such as IMO Resolution MSC.428(98) (IMO, 2017a). The plan includes various procedures related to risk management, threat and vulnerability assessment, access

Table 3
Learner feedback on cyber security practices by operators.

Question	Operator A	Operator B
Would you like your operator to provide dedicated training on the cyber security policies and procedures within the SMS?	yes: 100%	yes: 75% no: 13% unsure 13%
The online maritime cyber security training provided by your operator is repeated once every [Operator A: five/Operator B: two] years. Do you think this interval is sufficient?	no need: 13% sufficient: 38% each year: 38% every 3 years: 13%	each year: 100%
What is your general opinion about the online maritime cyber security training provided by your operator?	great: 50% good: 38% average: 13%	great: 50% good: 25% average: 13% poor: 13% yes: 100%
Do you think your operator should allocate more resources to its cyber security needs?	yes: 63% no: 13% unsure: 25%	yes: 100%
Would you like your operator to continue conducting ship-shore combined drills that cover different cyber attack scenarios?	yes: 100%	yes: 100%

control, asset management, incident response, and mobile device usage. The importance of training is also emphasised to improve the cyber awareness of the crew.

An SMS form is used to report, analyse, and address cyber security-related non-conformities and near misses onboard ships or at the operator's office. The form includes sections for describing the deficiency, immediate actions taken, violated procedures, and root cause analysis covering both technical and human element factors. It also requires the identification of corrective and preventive actions.

Another SMS form is used to document, investigate, and resolve cyber incidents onboard ships or at the operator's office. The form is completed to collect detailed information regarding an incident, such as the date, time, position, department, compromised systems, type of cyber incident (e.g. malware attack or data theft), and relevant personnel. The form also requires investigators to describe the sequence of events, root causes, corrective and preventive actions, and dissemination of lessons learned to crew members, shore-based staff, and external parties.

An emergency response procedure within the SMS provides instructions for GNSS failures. Although this procedure is primarily intended for use during GNSS failures, it also includes recommendations published by the Nautical Institute to detect potential GNSS jamming and spoofing (Gahnström, 2026).

According to the training matrix in the SMS, cyber security training for ship crew and shore-based staff is delivered online in their native language. The training takes approximately one hour and is refreshed every five years. As of the date of this study, no cyber security drill had been conducted onboard ships or as a ship-shore combined drill. However, for the first time, the operator had included a ship-shore combined drill based on a cyber security scenario in its annual drill plan to improve cyber awareness. At the time of this study, this planned drill had not yet been conducted. The ship-shore combined drill conducted within the scope of this study was the same drill that had been included in the operator's annual drill plan. Therefore, the drill conducted as part of this study also fulfilled the operator's planned annual drill.

Although the cyber security plan was recently revised, it still refers to several outdated cyber security-related documents issued by recognised organisations, such as the IMO and the International Organization for Standardization (ISO). Furthermore, although numerous procedures are documented in the plan, many of them are not implemented in practice, including asset management, vulnerability management, and risk management procedures.

Risk management has been a mandatory requirement for shipping companies since 2 January 2021 and is verified by flag state representatives (typically classification societies) onboard ships and at company offices during the renewal audits of the Document of Compliance (DOC) and Safety Management Certificate (SMC) (IMO, 2017b). Cyber risk management is also a requirement under the SIRE 2.0 (OCIMF, 2022) and CDI (CDI, 2023) programmes and is verified during onboard inspections. However, cyber risks are neither identified nor assessed. This significant deficiency has not been identified by either internal personnel or external auditors.

The IMO recommends that ship operators reference cyber risk management procedures from the SMS within the SSP (IMO, 2021). However, the SSP of Operator A contains no statements regarding cyber security.

5.2. Operator B

The cyber security policy of Operator B defines the overall approach for maintaining the cyber security of the operator's digital infrastructure and information systems. To this end, the policy identifies the purpose and scope of cyber security management, and the responsibilities of relevant personnel. It emphasises staff awareness, reporting obligations, and continuous monitoring to ensure effective cyber security.

A risk assessment is available for cyber risks onboard ships and at the

operator's office. Cyber risks are assessed using a conventional likelihood-severity matrix. A security breach affecting a vessel or the operator's office was identified as a potential cyber risk. According to the risk assessment, this risk may result in high-cost damage. Various mitigation measures are proposed, such as publishing bulletins, taking backups, and restricting network access. Following the implementation of mitigation measures, the risk level decreases from very high to medium.

The operator's SMS includes a step-by-step checklist to be followed in the event of a cyber incident onboard ships. At least one officer is assigned responsibility for implementing each step in the checklist. The checklist includes immediate actions, such as informing the master, sounding the general alarm, and notifying the Company Security Officer (CSO) and the CCySO. It also contains technical measures, such as shutting down the network system, preparing backups for installation, and disconnecting the compromised system from the automation control system.

The SSP briefly addresses cyber security. It states that virtual security threats, such as cyber attacks targeting electronic hardware and software, may disable vessel operating systems and that measures against such threats are defined in the operator's cyber security plan. In addition, it notes that many vessel control systems can be operated remotely and may therefore be vulnerable to cyber attacks. However, these systems are not specified.

An SMS form is used to document newly joined crew members onboard fleet ships. According to the procedure, each crew member must receive cyber security familiarisation training within 24 h of joining or before the vessel sails. The form includes training subjects such as the responsibilities of the SCySO, the Wireless Fidelity (Wi-Fi) policy, the location of the cyber security plan onboard, and rules regarding the use of personal devices. The form is signed by the seafarer, the SSO, and the master. Based on our observations, the form appears to have been adapted from (Witherbys, 2025) and subsequently integrated into the operator's SMS.

The operator published a poster to be displayed on the bridge, in the engine control room, and in the cargo control room (CCR) to improve the cyber awareness of crew members. The poster provides various recommendations, such as creating complex passwords, backing up important files, and opening e-mails carefully sent by unknown senders. Moreover, it is recommended to contact the IT department in the event of suspicious activity on computers. The contact details of the IT department are also displayed on the poster.

Another published poster identifies the designated personnel responsible for supporting and managing shipboard IT and OT systems. The poster includes the names and contact details (telephone numbers and e-mail addresses) of the responsible personnel (i.e. CCySO). The poster instructs crew members to contact these individuals if assistance is required regarding shipboard IT or OT systems. It also specifies that the notice must be displayed onboard, including on the bridge, in the CCR, captain's office, ship's office, and the engine control room.

Each office staff member and crew member receives basic cyber security training. In addition, shore-based personnel and onboard officers receive advanced cyber security training. Each training course takes approximately one hour, and the training sessions are refreshed every two years. According to the annual drill plan, each ship must conduct a cyber security drill annually. 2nd Officers onboard ships are assigned the role of SCySO because they are responsible for the navigation of the ship.

Cyber security plans must be operator-specific. Reference books and guidelines may only serve as supportive elements in the development of such plans. The dedicated cyber security plan of Operator B appears to have been directly copied from an outdated guideline (BIMCO, 2021) (current version: (BIMCO, 2024)). The shore-based staff was not even aware of this issue. Similar to the deficiency identified in Operator A (Section 5.1), this deficiency in Operator B has also not been identified by either internal personnel or external auditors.

In addition to the dedicated cyber security plan, the SMS manual also

includes a section on cyber security. However, this section appears to have been directly copied from (Scribd, 2026) and remains considerably brief.

A cyber risk assessment is available within the SMS. However, it includes only a single risk, which is related solely to IT systems. Cyber risks affecting OT systems onboard ships are not considered. Furthermore, although a cyber attack scenario is planned annually for ship-board drills, no ship-shore combined drill is considered to evaluate the response of shore-based personnel to cyber incidents. Cyber security familiarisation training for newly joined crew members is delivered by the SSO instead of the SCySO.

6. Conducting a ship-shore combined drill and a lecture

Section 6 presents the implementation and findings of the ship-shore combined drills and lecture sessions conducted with the two tanker operators by following the phases of the MarCy training programme. It explains how the training was designed, conducted, and evaluated, and presents the key observations obtained regarding preparedness against GNSS spoofing incidents.

In this study, percentages rather than numbers were reported for the clarity and consistency of the survey results. During the analysis, it was observed that some learners skipped certain questions in the assessment survey. This resulted in variations in the total number of responses across items. Consequently, percentages were calculated based on the number of valid responses for each question. As a result of rounding decimal values to whole numbers, the aggregated percentages for some questions occasionally sum to slightly above or below 100%. This reflects a common statistical artefact of rounding and does not indicate any inconsistency in the data.

6.1. Identify the needs of the organization

The purpose of this stage is to identify the nature of the problem (Oruc et al., 2024a). To this end, the problems and expectations of the operators were discussed during the planning meetings attended by the operators' representatives listed in Table 4.

Both operators accepted our invitation for collaboration because of the various advantages such training could provide. Operator A operates nine oil/chemical tankers, whereas Operator B operates 13 oil/chemical tankers under the SIRE 2.0, CDI, and TMSA programmes. Therefore, both operators are required to conduct ship-shore combined drills under the ISPS Code and vetting programme requirements, as discussed in Section 2.2. Furthermore, as also stated in Section 2.2, this training could assist the operators in meeting the cyber security training requirements of the vetting programmes. Lastly, the operators intended to assess their cyber security capabilities and improve the cyber awareness of both the ship crew and shore-based staff.

Both operators were given complete freedom regarding the selection of the drill scenario. In other words, any potential drill scenarios were not suggested to the operators in advance. Operator A operates its tankers in the Black Sea and the Mediterranean. Operator B operates its tankers in the Red Sea and the Black Sea. Tankers of both operators frequently encounter GNSS spoofing attacks in these regions. Therefore, the operators needed to improve cyber awareness for GNSS spoofing in

particular.

The operators also assumed that ship crews were generally familiar with GNSS spoofing. However, they were uncertain about the crews' ability to respond effectively during an incident and follow the procedures during the response process. They were also unsure whether the crews were sufficiently familiar with the relevant materials included in the operators' SMSs and whether the existing procedures themselves were adequate.

6.2. Specify job performance

The objective of this stage is to understand the roles and responsibilities of potential learners (Oruc et al., 2024a). During the planning meetings, neither of the ship operators could provide an exact learner list for the training. Because the employees might have unscheduled visits or unexpected urgent tasks during the day. Therefore, information was obtained at the planning meetings about the roles and responsibilities of the potential learners instead of the exact learners.

The assessment survey conducted during the training also included questions regarding the professional backgrounds of the learners. Based on the responses provided in the assessment survey, the roles and responsibilities of the learners involved in the training sessions are presented in Table 5. Some learners had multiple roles, such as Chief Officer/SSO. The expectations provided by the operators' representatives during the planning meetings were largely consistent with the actual learners profiles presented in the table.

6.3. Identify learner needs

The purpose of this stage is to understand the specific needs of potential learners by considering their responsibilities (Oruc et al., 2024a). According to the MarCy training programme, the responsibilities of potential learners identified through the operators' representatives in Section 6.2 should normally be matched with the relevant training modules presented in Table 1. However, this training was limited to the cyber risks of GNSS technology onboard ships, which was only a specific component of Module 5: Critical Deck Systems. Since all learners would attend the same training session, it was unnecessary to match the responsibilities of potential learners with training modules.

As discussed in Section 6.5, the training curriculum should have been designed by considering the learners' backgrounds and responsibilities. During the planning meetings, it was anticipated that learners with both maritime and non-maritime backgrounds would participate in the training. The expectations of the operators' representatives were confirmed during the training sessions. Based on the responses provided in the assessment survey, the backgrounds of the learners involved in the training sessions are presented in Table 6.

In the assessment survey, learners were also asked about the relevance of the training to their job responsibilities. All learners from Operator A and 75% of those from Operator B indicated that the training was related to their responsibilities.

In the initial study, the MarCy training programme proposed training modules for each responsibility (Oruc et al., 2024a). In a subsequent study involving 79 learners, these proposed mappings were revised based on the findings obtained (Oruc et al., 2024b). In this study, the accuracy of these revised module-responsibility recommendations was evaluated once again.

To this end, only the responses of learners with a single responsibility were considered regarding the relevance of the training to their responsibilities. Responses from learners with multiple responsibilities were excluded, since their feedback could reflect the interaction of several responsibilities. This could obscure the direct relationship between specific responsibilities and the corresponding training modules. In contrast, responses from learners with a single responsibility provide a more precise and unambiguous representation of this relationship.

Based on the findings of the assessment survey, Table 7 shows

Table 4
Representatives of the operators at the planning meeting.

Operator A	Operator B
DPA/CSO/HSQE Manager	DPA/CSO/HSEQ Manager
Training Manager	
Fleet Manager	
IT Manager	
DPA: Designated Person Ashore CSO: Company Security Officer	
HSQE: Health, Safety, Quality, Environment	
HSEQ: Health, Safety, Environment, Quality	

Table 5
Roles and responsibilities of learners.

Operator A or B	Learners' Roles	Learners' Responsibilities							
		training activities	cyber security activities	IT activities	investments and management for marine operations	marine operations	safe manning of ships	navigation, communication, and cargo operations	maintenance and operation of the ship's main propulsion and auxiliary systems
A	Master							✓	
A	2nd officer/SCySO		✓					✓	
A	Chief engineer								✓
A	Fleet manager	✓			✓	✓			
A	DPA/CSO/HSQE manager				✓				
A	Training manager	✓							
A	IT manager		✓	✓					
A	Marine/HSQE superintendent	✓							
B	Master							✓	
B	Chief officer/SSO							✓	
B	3rd officer							✓	
B	Chief engineer								✓
B	DPA/CSO/HSEQ manager				✓				
B	Crew manager	✓					✓		
B	Crew senior assistant manager						✓		
B	Crew assistant manager						✓		
B	Marine/Training superintendent	✓				✓			
B	IT manager			✓					
B	Marine superintendent					✓			
B	Senior operation specialist					✓			

Table 6
Backgrounds of learners.

Background	Operator A	Operator B
deck	5	7
marine engineering	1	1
marine and port management	1	
computer science	1	1
maritime business management		1
business administration		1
economics		1

Table 7
Relevance of training to the responsibilities of learners.

Responsibility	Op. A	Operator B		
	Yes	Yes	No	Unsure
training activities	1			
cyber security activities				
IT activities			1	
investment and management of marine operations	1	1		
marine operations		2		
safe manning of ships			1	1
navigation, communication, and cargo operations	1	3		
maintenance and operation of the ship's main propulsion and auxiliary systems	1	1		
Op. A: Operator A				

whether learners with a single responsibility perceived the training as relevant to their specific job duties.

Two learners responsible for Safe Manning of Ships considered that the training might not be relevant to their duties. In our previous study (Oruc et al., 2024b), we also stated that Module 5: Critical Deck Systems was not necessary for this responsibility.

The learner responsible for IT activities works as the IT manager in Operator B. He considered the training unrelated to his job responsibilities. As seen in Table 5, this learner did not select cyber security responsibility for his role in the assessment survey. However, as presented in Sections 5 and 5.2, according to the procedures, the IT manager is designated as the CCySO. Based on these procedures, he is responsible for the cyber security of both IT and OT systems. This information was also confirmed by the representative of Operator B at the planning meeting. Considering these observations, it is possible that the IT manager was not aware of his cyber security responsibilities. On the other hand, the IT manager of Operator A was not included in Table 7, as he was responsible for both cyber security and IT activities. Nevertheless, he considered the training useful for his own responsibilities.

The responses provided by the learners in the assessment survey supported the accuracy and adequacy of the recommendations proposed for this step of the MarCy training programme in our previous study.

6.4. Determine objectives

The objective of this stage is to define the specific objective and learning outcomes for each training module (Oruc et al., 2024a). Table 8 presents the specific objective and learning outcomes for Module 5: Critical Deck Systems, which already covers GNSS vulnerabilities. Therefore, there was no need to revise the defined objective and learning outcomes of Module 5 within the MarCy training programme.

Table 8

The objective and desired learning outcomes of the M5 critical deck systems (Oruc et al., 2024a).

Objective	Desired Learning Outcomes
Learners will learn cyber risks in the deck systems.	- Learners will acquire a comprehensive knowledge of cyber risks related to critical deck systems and will be proficient in implementing appropriate mitigation measures. - Learners will be capable of effectively handling cyber security incidents related to critical deck systems

However, we slightly refined them to define the specific objective and learning outcomes for the training, as shown in Table 9. This enabled us to better understand how to design more targeted training in line with the expectations of the operators.

6.5. Build curriculum

The objective of this stage is to identify a curriculum to meet the desired objective and learning outcomes represented in Table 9 (Oruc et al., 2024a). The curriculum includes two groups, such as essential and helpful content (Nadler and Nadler, 1982). Essential content is required to achieve the learning objectives and outcomes. Helpful content is supplementary and provides a foundation for understanding the essential content.

The curriculum was designed by considering the responsibilities and backgrounds of potential learners. As presented in Table 10, a content list was developed to achieve the defined learning objective and outcomes by following a general-to-specific approach. Moreover, the potential materials that could be used as training resources were also identified at this stage. Ship and operator-specific materials were provided for training sessions by the operators.

The curriculum represented in Table 10 was specifically designed for the lecture part of the training. After the drill, the foundation information about GNSS technology was delivered to the learners. Afterwards, cyber security-related topics were introduced. Last, ship and operator-specific subjects were explained.

As explained in Section 6.3, it was understood during the planning meetings that the training sessions would include learners from both maritime and non-maritime backgrounds across the two operators. Therefore, some of the helpful content included in the curriculum was not directly related to cyber security, but was instead intended to provide foundational maritime knowledge for learners with non-maritime backgrounds. In particular, explaining GNSS technology was considered necessary to support their understanding of the training content.

Ship-specific content particularly included systems on the bridge and in the engine room that depend on information transmitted from the GNSS receiver. Therefore, the learners would become aware of all on-board systems that could be affected by a potential GNSS attack. Such information was obtained from the operators' representatives during the planning meetings.

During the planning meetings, all documents related to GNSS and

Table 9

The desired objective and learning outcomes for the training.

Objective	Desired Learning Outcomes
Learners will learn cyber risks of the GNSS.	- Learners will acquire a comprehensive knowledge of cyber risks of GNSS and will be proficient in implementing appropriate mitigation measures. - Learners will be capable of effectively handling cyber security incidents related to GNSS. - Learners will be familiar with company procedures regarding cyber security and GNSS risks. - Learners will be aware of the impact of GNSS risks as ship-specific.

Table 10

Curriculum and useful materials.

Content	E/ H	Material
GNSS technology and its variants (i.e. GPS, GLONASS, Galileo, and BeiDou)	H	(Oruc et al., 2022)
features of a GNSS receiver (including Multi-GNSS)	H	(Oruc et al., 2022)
features of the D-GNSS and SBAS	H	(Oruc et al., 2022)
GNSS jamming and spoofing	E	(Oruc et al., 2025)
threat actors	E	(Oruc, 2020)
impact factors of GNSS attacks	E	(Glomsvoll and Bonenberg, 2017)
incidents experienced	E	(M-CERT, 2026; Stenden, 2026)
comparison of GNSS jamming and spoofing	E	(Warner and Johnston, 2003)
attack response	E	(Oruc et al., 2025)
monitoring services of GNSS attacks	E	(GPSwise, 2026; Wiseman, 2026)
reporting channels for GNSS attacks	E	(NATO Shipping Centre, 2026; USCG Navigation Center, 2026)
ship-specific topics	E	operator
operator-specific procedures	E	operator
E: essential content H: helpful content		

cyber security within the SMSs of both operators were obtained. These materials were specifically discussed during the training sessions with each learner group. The relevant documents are explained in Section 5.

The MarCy training programme proposes prerequisite modules. Module 1: Basic Cyber Security should be completed before enrolling in any other module. However, all learners had already received cyber security training, as discussed in Section 5. Therefore, during the planning meetings, both operators stated that there was no need to deliver Module 1. Consequently, this module was not delivered to the learners.

6.6. Select instructional strategies

The purpose of this stage is to determine the most appropriate instructional strategies for the identified curriculum (Oruc et al., 2024a). The MarCy training programme suggests five instructional strategies for maritime cyber security training: drill, lecture, discussion, case study, and demonstration.

At the beginning of the planning meetings, a ship-shore combined drill was planned as the primary instructional strategy. However, following the discussions, it was understood that a lecture component would also be required. During the drill, learners could make various mistakes in the implementation of the procedures. Moreover, their awareness regarding the features and cyber vulnerabilities of GNSS needed to be improved. Considering these expectations, the training was designed by combining drill and lecture strategies.

During the drill, learners had the opportunity to discuss the incident internally and with representatives of third-party stakeholders. Questions directed to the author of this study were not accepted during the drill in order to preserve the realism of the scenario. Nevertheless, learners raised various questions among themselves and during discussions with third-party stakeholders. Furthermore, discussions continued internally and with the author following the drill. Although a structured discussion format was not formally implemented in the training, elements of discussion-based learning were partially incorporated.

At the end of the training, learners were asked about the effectiveness of the instructional strategies through the assessment survey. Moreover, their opinions regarding the instructional strategies that were not implemented in the training (i.e. discussion, case study, and demonstration) were also collected. Table 11 summarises the learners' views on the instructional strategies.

A total of 20 learners, including shore-based staff and seafarers, responded to the relevant questions in the assessment survey. The implemented drill and lecture strategies were considered effective by the learners. According to their responses, the instructional strategies that were not implemented in the training, namely discussion, case

Table 11
Learners' opinions on instructional strategies.

Instructional Strategy	Operator A		Operator B		Overall	
	E	I	E	I	number	rate
drill	8	-	12	-	20	100%
lecture	8	-	12	-	20	100%
discussion	-	5	-	12	17	85%
case study	-	5	-	12	17	85%
demonstration	-	5	-	12	17	85%
	100%	62.5%	100%	100%		

E: It was effective. | I: It should have been implemented.

study, and demonstration, should also have been incorporated.

6.7. Obtain instructional resources

The purpose of this stage is to ensure the availability of all resources required for cyber security training (Oruc et al., 2024a). This section addresses physical and human resources, as well as the training materials.

A PowerPoint presentation was prepared for the lecture component of the training. The presentation content comprised books, scientific papers, guidelines, circulars, videos, images, and photographs. Ideally, this presentation should have been shared with the learners before the training. However, doing so could have revealed the drill scenario to the learners in advance. Consequently, the learners might have had the opportunity to familiarise themselves with the relevant company procedures before the drill.

Drills are conducted not only for training purposes but also to evaluate the existing knowledge and preparedness of personnel. Therefore, the ship crew and shore-based staff should not be informed about the drill scenario beforehand.

The master of the ship managed by Operator A commented that prior information regarding the drill scenario would have been helpful, as the crew could have prepared accordingly. However, the operator's representative emphasised that the purpose of such drills is also to assess the crew's level of awareness and the operator's ability to identify potential weaknesses. The representative further pointed out that, in the event of an actual cyber incident, there would be no prior warning. Consequently, the drill scenario should not be disclosed in advance to the ship crew, shore-based staff, or third-party stakeholders in order to obtain the most realistic outcomes.

All learners and the lecturer spoke the same native language. Therefore, the lecture materials and the assessment survey were prepared in that language. During the training, all communication was also conducted in the native language. However, various maritime and cyber security terms were retained in English, as using local translations of certain terms (e.g. AIS) could have made understanding more difficult for the learners.

During the lecture, various questions beyond GNSS spoofing were asked to the lecturer, such as cyber security-related clauses in marine insurance policies (e.g. Protection and Indemnity (P&I) and Hull and Machinery (H&M)). This situation revealed that the lecturer needed to have full mastery of maritime cyber security. The study also highlighted the importance of qualified human resources for the effective delivery of such training.

During the training, the ship crew of both operators was on the bridge. The office staff was in the emergency response center of the operators. Both centers had reliable infrastructure, including ventilation, electricity, lighting, and broadband internet. Moreover, they were equipped with speakers, large screens, professional webcams, and microphones.

All communication between the office, ships, and third-party stakeholders was conducted through WhatsApp, mobile phones, and e-mail. One day before the training, the MS Teams connection between the ships and the emergency response center of the operators was arranged and

tested by the operators' IT departments. However, MS Teams was not used to conduct the drill itself. Instead, it was used to display the PowerPoint presentation and observe discussions among ship crew members on the bridge during the drill. Furthermore, the training sessions needed to be recorded for the analysis conducted in this study and for the operators' internal evaluations. Both operators also intended to use these training recordings for future internal training purposes.

One hour before the training sessions, the lecturer was present at the emergency response center of both operators to identify and prevent potential problems that might arise during the training. The IT Manager of Operator A had initially established the MS Teams connection in the CCR of the ship. However, navigation systems are on the bridge, and the GNSS spoofing drill scenario was to be conducted there. Therefore, the ship crew needed to remain on the bridge during the drill. Although the IT Manager had participated in the planning meetings, this issue had not been foreseen. Following the lecturer's notice, the MS Teams connection was re-established on the bridge before the training began.

The assessment survey was developed by incorporating feedback received from the operators. One day before the training, the final version of the assessment survey was shared with both operators for printing. However, Operator B unintentionally printed the draft version instead of the final version. This issue was identified by the lecturer and corrected before the training began, and the final assessment survey was subsequently printed.

The issues experienced during the preparation process highlighted the importance of conducting checks before the training sessions. They also revealed that IT departments may not be sufficiently familiar with marine operations. Therefore, clear instructions should be provided to them in advance. Moreover, assessment surveys should preferably be prepared and kept ready by the lecturers rather than by the ship operators.

6.8. Conduct training

The objective of this stage is to conduct the planned training, including both a ship-shore combined drill and a lecture (Oruc et al., 2024a).

The ship crew and shore-based staff were informed of the training date and time two days in advance in order to arrange their work schedules accordingly. To ensure realism, the drill scenario was not shared with any personnel (except for employees in the planning meeting) to prevent advance preparation.

The assessment survey was sent to the ships via e-mail 15 min before the training began. It was then printed and distributed to the learners onboard. In contrast, the assessment survey for the shore-based personnel was distributed at the beginning of the training. In this way, reviewing the assessment survey in advance did not allow learners to acquire prior knowledge about the planned drill.

The training commenced with a 10-minute opening speech. During the opening speech, the representatives of the operators and the lecturer explained the purpose of the training and the study. Moreover, the structure of the training and the assessment survey were discussed. Subsequently, the first section of the assessment form was completed by the learners.

The training lasted approximately two hours for each operator. One hour consisted of the ship-shore combined drill, while the other hour comprised the lecture component of the training. During the lecture, learners' questions were also answered. Moreover, maritime cyber security incidents, particularly GNSS spoofing incidents, were discussed. As seen in Table 12, all learners stated that they had the opportunity to ask questions during the training. The vast majority of learners expressed that the duration of the training was sufficient. The feedback collected from the learners indicated that two hours of training was sufficient for GNSS spoofing response training.

As stated in Section 2.2, the ISPS Code requires the involvement of third parties in ship-shore combined drills. This enables the assessment

Table 12
Interaction opportunity and training time evaluation.

Questions	Operator A		Operator B		
	Yes	Unsure	Yes	No	Unsure
Did you have the chance to ask questions during/at the end of the training?	100%		100%		
Was the allocated duration of the training sufficient?	75%	25%	83%	8%	8%

of third-party companies' competencies in responding to incidents and allows potential communication issues to be identified. The third-party organisations involved in the drills conducted by Operator A and Operator B are presented in Table 13.

Once the drill commenced, these stakeholders were notified via e-mail, with it being clearly stated that the situation was part of a drill scenario, in order to simulate a realistic incident response process. In case of an incident, operators prefer to initiate communication via e-mail rather than by phone. This is because such correspondence may later serve as evidence during incident investigations conducted by authorities such as insurance companies, classification societies, or flag states. Phone communication may subsequently follow if necessary.

In Table 13, the term *general* refers to general e-mail addresses of third-party organisations (e.g. info@company.com). The term *individual* refers to individual corporate e-mail addresses assigned to employees within third-party organisations (e.g. firstname.lastname@company.com). The term *emergency* denotes dedicated e-mail addresses established by third parties specifically for emergency situations.

Communication was established with 10 different maritime stakeholders during the drill, including one marine insurer that participated in both drills. The initial notification was sent by the operators to the stakeholders' general, individual, or emergency e-mail addresses. Some of these stakeholders did not provide any response to the initial notification sent by the operators during the training.

During the drill, all written correspondence was maintained in English. Therefore, all information received from the ships could be forwarded to external drill participants when necessary. Moreover, English is identified as the working language of both operators in their SMSs. Thus, all correspondence was required to be conducted in English.

Following the drill, a lecture was delivered to all the learners. The lecture component of the training lasted approximately an hour for each operator. During the lecture, learners were able to ask questions directly to the lecturer and therefore did not need to wait until the end of the session. Discussions were conducted both among the learners and between the learners and the lecturer.

Table 13
Third-party responses to the drill notification.

Operator	3rd Party	e-Mail Type	e-Mail	Phone
A	classification society	general	X	-
A	local maritime authority	general	X	-
A	marine insurer	individual	X	✓
A	marine electronic service provider	general	✓	✓
A	marine insurer	general & individual	✓	✓
A	ship agency	general	✓	✓
B	classification society	individual	X	-
B	marine insurer	individual	✓	-
B	marine electronic service provider	individual	✓	✓
B	flag state	emergency	X	-
B	ship agency	general	✓	-

X: Not replied by 3rd party | ✓ Replied by 3rd Party | - Not contacted with 3rd party

6.8.1. Drill with operator A

According to the scenario, the vessel was navigating in the Black Sea on a route toward the Bosphorus Strait. The master called the DPA and reported that the GNSS receiver was displaying incorrect position data. He also stated that the position information obtained from the other GNSS receiver and the AIS onboard was consistent but incorrect. This situation raised the possibility of a GNSS spoofing attack.

The operator's Emergency Response Team was convened by the DPA in the Emergency Response Room. The third parties listed in Table 13 were notified by the DPA via e-mail. In addition, the superintendent and the DPA contacted the third parties by phone. Communication with the vessel was maintained via e-mail and voice calls over WhatsApp. The vessel submitted the Initial Report, as specified in the operator's SMS, including details of the vessel and the incident.

During the drill, nearby vessels were assumed to have been warned about the GNSS spoofing attack. The number of lookouts was increased on the bridge. The speed of the vessel was also reduced. The vessel was instructed to proceed using dead reckoning (DR mode) on the ECDIS in order to enter the RADAR coverage area of the Vessel Traffic Services (VTS), thereby enabling safe navigation with VTS support. Anchoring was also considered as an option, therefore, advice regarding anchorage areas was requested from the agent. However, the vessel's proximity to the coast provided an advantage, allowing it to continue safe navigation using its own RADAR without the need to anchor.

The vessel was equipped with a Multi-GNSS (M-GNSS) receiver supporting both GPS and GLONASS. The receiver was operating in automatic mode. Then, it was switched to manual mode, and the vessel's position was observed with GPS and GLONASS separately, at the suggestion of the service provider. Based on the scenario, it was assumed that GPS was targeted, and safe navigation subsequently continued using GLONASS.

All third parties were informed via e-mail that the drill had been completed. Following the drill, the Cyber Incident Report and Cyber Non-Conformity/Near Miss Report forms specified in the operator's SMS were completed.

6.8.2. Drill with operator B

According to the scenario, although the vessel was navigating in the Eastern Mediterranean, its GNSS coordinates falsely indicated that it was at a Russian port in the Black Sea. The officer of the watch identified the anomaly and informed the master. After verifying the positional discrepancy, the master contacted the operator's emergency hotline via satellite phone. The call was answered by the DPA, who received preliminary information from the vessel. Following this, an emergency meeting was convened in the emergency response room upon the DPA's request.

The ship's crew, without considering the possibility of a GNSS receiver malfunction, directly assumed that the incident was a cyber attack. Upon the suggestion of the office personnel, the position was cross-checked using another GNSS receiver on the bridge, as well as via AIS and Inmarsat-C, both of which had independent GNSS receivers. It was confirmed that all receivers were displaying the same, but incorrect, position data. Subsequently, the DPA sent a notification e-mail to third parties regarding the cyber incident.

Assuming that the attack might have been conducted via the internet, the CCySO (IT Manager) directed several questions to the vessel. The crew reported that no anomalies other than the GNSS position discrepancy had been observed, no new software had been installed recently, and no files received via e-mail had been opened. The service provider was contacted by phone to inquire whether there was any ongoing GNSS-related attack in the region and to seek recommendations. The provider stated that they were not aware of any such incident and could not offer specific advice.

The office staff advised the master to proceed using traditional navigation methods toward the Fethiye anchorage area. The bridge team was reinforced by assigning one officer, one lookout, and one helmsman,

and the vessel was directed toward the anchorage through the DR mode on the ECDIS. A ship agent was appointed. The voyage was estimated to take approximately 15 h. It was also decided that the relevant VTS station would be informed before the vessel entered the VTS coverage area.

It was planned to install GNSS attack detection equipment onboard at the Fethiye anchorage area. The vessel completed and submitted the operator's Risk Assessment form and Cyber Security Attack Checklist. Finally, an e-mail was sent to third parties indicating that the drill had been completed, thereby concluding the drill.

6.9. Evaluation and feedback

The purpose of the evaluation stage is to analyse the results and objectives achieved through the training programme (Oruc et al., 2024a).

During the training sessions, each learner completed two multiple-choice exams called Quiz and Test. The quizzes were completed before the training. The tests were completed after the training. These exams were conducted to evaluate the learners' knowledge levels before and after the training, thereby enabling the effectiveness of the training to be assessed. Approximately 10 min were allocated for the quiz and 15 min for the test in each training session. The quiz consisted of six questions. On the other hand, the test included ten questions. Both the quiz and the test covered three groups of topics: operator procedures, GNSS technology, and GNSS-related cyber risks. All six questions from the Quiz were repeated in the post-training Test, which also included four additional questions. To reduce the potential influence of question and answer-option order, the order of both questions and answer options was randomised in the post-training Test. This assessment approach was also used in the previous evaluation of the MarCy training programme (Oruc et al., 2024b).

Table 14 presents the mean scores and standard deviations (SDs) of learners in the Quiz and Test on a 100-point scale. The table also shows the increase in scores following the training. The mean Quiz score was 40.83 (SD = 17.50). The mean Test score was 59.50 (SD = 14.32). On average, the training sessions resulted in a 46% increase in learner scores. A paired-samples *t*-test indicated a statistically significant increase in learner scores following the training ($t(19) = 3.787$, $p = 0.001$). The mean increase was 18.67 percentage points (95% confidence interval [CI]: 8.35–28.98). A large effect size was observed (Cohen's $d_z = 0.847$). At the individual level, scores increased for 16 learners (80%) and decreased for four learners (20%). No learner showed an unchanged score. These results indicate that the increase in learner scores was statistically significant and of a large magnitude. The CI remained above zero. The individual-level results showed that 80% of the learners improved their scores.

The MarCy training programme does not recommend a specific score threshold for determining whether a learner is successful. Instead, training designers are expected to define their own success criteria based on their specific needs. Therefore, the increase in scores does not necessarily indicate whether the learners of the operators should be considered successful or unsuccessful. Although the training may be considered effective, some learners may still require repeated or additional training.

The primary users of GNSS are personnel with a deck background. As shown in Table 6, 60% of the learners had a deck background. By the end

of the training, these learners increased their scores by 17%, whereas learners from other backgrounds improved their scores by 109%. This finding indicates that the training was particularly effective for learners without navigation experience.

Learners were asked to evaluate the conducted training through score-based questions on a five-point scale. As shown in Table 15, the feedback from the learners of both operators was highly positive. All items received scores of 4.6 or higher. This indicates a high level of satisfaction with the overall effectiveness of the training.

The drill scenario was rated similarly highly by both operators. Moreover, the lecture delivered after the drill received a score of 4.9. It was perceived as highly useful. The most notable finding is that both operators rated the lecturer with a perfect score of 5.0, reflecting complete satisfaction with the lecturer's command of the subject.

To evaluate the training, learners were also asked a series of yes–no questions regarding its content, relevance, and clarity. As shown in Table 16, the results indicate that learners from both operators viewed the training positively. All respondents agreed that the training improved their understanding of GNSS-related threats.

A considerable proportion of learners, particularly from Operator A, indicated the need for additional maritime cyber security training beyond the cyber risks of GNSS. Overall, the presentation materials used in the lecture were considered clear. Most learners also felt that they had sufficient maritime knowledge to follow the training. These findings unveiled that the training was effective and highlighted a demand for broader maritime cyber security training.

Additionally, open-ended questions were asked to the learners in order to gain further insights into the training conducted. The learners indicated that maritime cyber security training should include OT-specific content such as GNSS-related risks. For this reason, the training was perceived as highly beneficial. Sharing websites that publish information on GNSS interference incidents was also considered useful during the training. On the other hand, although various animations and videos were used in the lecture, the learners suggested that their number could be further increased. This training was conducted on only one vessel from each operator, however, the learners stated that it should be delivered across all vessels in their fleets.

7. Discussions

This section discusses the findings obtained from the training sessions conducted with the two ship operators and evaluates their implications for maritime cyber security preparedness. First, observations from the drills and training activities are analysed. Then, the discussion examines the applicability and effectiveness of the MarCy training programme in light of the findings, followed by an assessment of the operators' procedures and supporting materials related to cyber security management.

7.1. Discussion on the training

This section discusses the observations made during the training sessions conducted with the two ship operators. The findings highlight how learners and stakeholders responded to the simulated GNSS spoofing incident, identify procedural and operational deficiencies, and

Table 14
Average quiz and test results by operator.

	Operator A	Operator B	Overall
Quiz	48	36	41
Test	68	54	60
Score Increase	20	18	19
Ratio Increase	41%	50%	46%

Table 15
Feedback about the training.

Question	Operator A	Operator B
Were you satisfied with the overall effectiveness of the training conducted?	4.6	4.8
Was the drill scenario presented interesting?	4.6	4.7
Was the lecture provided after the drill useful?	4.9	4.9
How was the lecturer's command of the subject?	5.0	5.0

Table 16

Cross-operator analysis of training-related perceptions.

Question	Operator A		Operator B		
	Yes	Unsure	Yes	No	Unsure
Did the training help you understand the main threats related to GNSS?	100%		100%		
Besides GNSS risks, do you believe you need further training on maritime cyber security?	75%	25%	67%	17%	17%
Was the presentation material easy to understand?	88%	13%	100%		
Was your maritime knowledge sufficient to understand this training?	88%	13%	92%		8%

evaluate the effectiveness of the operators' existing cyber security measures. The discussion also demonstrates how drills can reveal gaps that are difficult to identify through document reviews or theoretical training alone, thereby providing valuable input for improving procedures, asset management, communication practices, and future cyber security training.

7.1.1. Training with operator A

Operator A sought to determine whether the detected position discrepancy was caused by a malfunction or a cyber attack. To this end, the officers onboard compared the position data obtained from another GNSS receiver and from the AIS. According to the scenario, the vessel's position appeared identical across all devices, although the displayed position was incorrect. This is an expected condition during a GNSS spoofing attack, as all receivers within the affected area are typically influenced.

The service provider was asked whether they had any information regarding GNSS-related attacks in the area where the vessel was operating. The service provider indicated that GNSS attacks had occasionally been reported in the region. Operator A could also have verified this information through the GPSJAM (Wiseman, 2026) and GPSwise (GPSwise, 2026) platforms, however, they were not aware of these sources.

An e-mail was sent to the individual corporate e-mail address of the contact person at the marine insurer. However, when no response was received, the insurer was contacted by phone. Then, it was learned that the individual was on leave. In urgent situations, generic or dedicated emergency e-mail addresses should be used instead of individual corporate e-mail accounts. Relying solely on an individual corporate e-mail address in such situations constitutes a communication weakness. The insurance company, in turn, should clearly inform its clients of the appropriate emergency contact e-mail address. Furthermore, personnel within the insurance company should anticipate that clients may need to reach them in urgent situations and therefore configure automatic out-of-office replies when they are on leave.

The master of the ship delayed the preparation of the operator's Initial Report form and made an error by omitting essential information such as the vessel's route and speed. Following a warning from the superintendent, the form was corrected and resubmitted. The late submission of the form caused a delay in the operator's situational awareness. This situation highlighted the need for further training for the ship's master.

During the drill, a simulated warning was issued to nearby vessels regarding a GNSS spoofing attack. However, it was not inquired whether they were experiencing a similar issue. Since GNSS attacks are typically region-specific, confirmation could have been obtained from other vessels operating in the same area. This would have helped determine whether the issue resulted from a GNSS receiver malfunction or a GNSS spoofing attack.

The operator has a GNSS Failure Checklist within its SMS. This

checklist should have been followed from the very beginning of the incident. However, the items included in the checklist were reviewed only midway through the drill.

The GNSS variant used onboard was not known. After consulting the service provider, it was determined that both GPS and GLONASS were in use. This revealed deficiencies in the operator's asset management practices.

According to the procedures, in the event of a cyber attack, the operator's IT manager, who also serves as the CCySO, should be contacted. However, the vessel contacted the DPA instead. It was discussed that, in the case of attacks targeting OT systems, the IT manager's expertise alone might be insufficient, therefore, this procedure should be reviewed.

7.1.2. Training with operator B

During the drill, Operator B's routine operations e-mail address (safety@operatorB.com) was kept in cc throughout the correspondence. As a result, e-mails related to the operations of other vessels in the fleet were also received by members of the emergency response team. In the event of an actual incident, all responsible personnel should focus solely on the ongoing situation. Therefore, as implemented by Operator A, the use of a dedicated emergency e-mail group would be beneficial.

At the beginning of the drill, both the ship crew and office-based personnel followed the Cyber Security Attack Checklist in the SMS of the operator. However, during the drill, they discontinued following the checklist. The checklist did not contain sufficient instructions for responding directly to GNSS attacks, and this deficiency was identified during the drill. Therefore, the checklist should be revised by considering potential cyber attacks, including GNSS-related incidents.

In addition, errors were observed in the implementation of the checklist. The checklist includes separate sections for cyber attacks targeting systems such as networks, machinery systems, and cargo systems. During the GNSS attack scenario, shipboard and shore-based personnel should have followed the navigation-related section of the checklist. Instead, they proceeded sequentially according to the numbering of the checklist and began with the checks listed under the network section. This situation revealed a lack of training among the personnel. Therefore, both shipboard and shore-based personnel should be trained on the proper implementation of operator procedures.

By following the instructions in the network section of the checklist, the learners attempted to respond to the simulated GNSS spoofing attack. As a result, the vessel's internet connection was disconnected in accordance with the procedures. (Since the drill was being monitored via Microsoft Teams, the connection was assumed to be terminated.) However, GNSS spoofing attacks do not inherently occur through internet-based communications. The learners were not aware of this distinction. This situation demonstrated that the learners require specific training on cyber attacks targeting onboard OT systems and on the characteristics of different maritime cyber attack types.

During the simulated GNSS spoofing scenario, the vessel was directed to an anchorage area approximately 15 h away. Since this area was within the coverage of the VTS, it was considered both a safe location for navigation and suitable for the service provider to install the GNSS attack detection system onboard the vessel. Although the operator could theoretically decide to divert the vessel to a distant area, such a measure may not be practically feasible due to commercial pressures.

It was observed that the operator has begun equipping its fleet with GNSS attack detection systems. However, no clear assessment regarding the effectiveness of these systems could be obtained from representatives of the operator.

7.2. Discussion on MarCy training programme

Since the cyber risks of GNSS technology are included as part of the content of "Module 5: Critical Deck Systems" in the MarCy training programme, there was no need to develop a new training module for this

study. The training delivered to the learners covered only a limited portion of the content of Module 5.

The MarCy training programme was evaluated based on the opinions of 19 experts. However, following the evaluation with 79 learners, almost all stages of MarCy were revised. This study confirmed the appropriateness of these revisions. Furthermore, it demonstrated that expert opinion alone may not be sufficient. Field findings and observations can lead to more accurate and reliable outcomes.

In the first study on the MarCy training programme (Oruc et al., 2024a), various responsibilities were introduced for ship crew and shore-based staff within ship operators. However, during the evaluation of MarCy conducted with learners (Oruc et al., 2024b), it was observed that these responsibilities were insufficient, and several recommendations for revision were proposed. This study confirmed that the revisions suggested in the previous study (Oruc et al., 2024b) were both accurate and sufficient.

As shown in Table 5, and consistent with the findings of previous studies based on the MarCy training programme, it was observed that various roles and responsibilities were combined within both Operator A and Operator B (e.g. Marine/Training Superintendent). This observation from the two participating operators indicates that the organisational structures of ship operators may vary and reflects differences in the allocation of roles and responsibilities.

In the previous evaluation of MarCy (Oruc et al., 2024b), lecture, discussion, and case study instructional strategies were implemented. Before the training, these strategies were not sufficiently supported by the learners. However, after the training, all learners considered them to be effective. In other words, the learners' opinions changed following the training. Moreover, the drill and demonstration strategies were not sufficiently supported in the previous evaluation, even though they had not been implemented. In this study, a drill approach was tested for the first time, and all learners were satisfied, as shown in Table 11. This finding demonstrated that learners may have preconceived biases regarding an instructional strategy before experiencing it. As a result, the lecture, discussion, case study, and drill instructional strategies have now been evaluated, and all have been found to be effective. However, the effectiveness of the demonstration instructional strategy recommended by the MarCy training programme has not yet been evaluated from the learners' perspective.

The MarCy training programme recommends notifying learners at least seven days before the training. According to the assessment survey results, all learners from both operators considered the two-day notification period to be sufficient. The total duration of the training in this study was approximately two hours per session, however, in our previous study, the training sessions lasted up to a full working day. Considering the findings of both studies, we suggest that longer training sessions may require a one-week notification period, whereas two days may be sufficient for shorter sessions.

The drills provided direct observations of learners' responses to the simulated GNSS spoofing incidents, including their implementation of procedures and coordination with relevant stakeholders. However, the MarCy training programme does not prescribe specific success criteria for determining whether a learner has successfully completed the training, as clearly stated in our previous MarCy evaluation article (Oruc et al., 2024b). Training designers should therefore define success criteria that reflect their organisation's specific needs and priorities. Accordingly, the present study does not classify learners as successful or unsuccessful. However, the assessment scores, learner feedback, and observations obtained during the drills support the effectiveness of the training.

Training designers can establish success and competency criteria according to their specific training objectives and needs. Such criteria could include incident recognition, response time, implementation of relevant procedures, and other role-specific performance indicators. Future studies of the MarCy training programme could focus on developing measurable competency and success criteria for its training

sessions.

Many maritime terms are used in English. For instance, although GPS is an English abbreviation, it is referred to by the same term among seafarers regardless of their native language. During the drill, seafarers communicated in their native language. Moreover, masters and officers naturally moved around the bridge during the exercise and were occasionally positioned far from the microphone. For these reasons, MS Teams was not always able to generate accurate transcriptions. This occasionally resulted in insufficient written records for Thematic Data Analysis. The issue was addressed by reviewing the video recordings. For similar studies in the future, the use of lapel microphones, particularly for seafarers participating in drills, may improve the quality and accuracy of recorded communications.

7.3. Discussion on operators' materials

This study does not include a systematic gap analysis of the cyber security procedures of the operators. Therefore, the accuracy of the procedures or the effectiveness of their practices is not comprehensively evaluated in this study. However, based on both our observations during the conducted drill and the limited assessment of the procedures, it appears that both operators may have significant deficiencies in the implementation of their procedures.

As discussed in Section 5.1, cyber risk management has been mandatory for ship operators for many years under IMO and vetting programme requirements. Moreover, as discussed in Section 5.2, cyber security plans must be operator-specific. These requirements had been verified numerous times onboard ships and at operators' offices by different recognised organisations and internal personnel, however, the identified deficiencies had not been recognised. Following notification by the lecturer of this study regarding these deficiencies, corrective actions were initiated by both operators to address them. However, this situation revealed that the inspections are not effective in verifying even essential cyber security requirements.

The primary duty of IT specialists is the management of IT systems, and cyber security is generally not among their primary responsibilities. However, ship operators may assign cyber security responsibility to IT specialists within their SMS. According to our observations in this study, the IT specialists in both operators appeared to have a limited understanding of the OT systems onboard ships and their relationship with cyber security. In fact, it was revealed that the IT manager of Operator B might not even have been aware of his OT cyber security responsibilities, despite the operator's own procedures.

8. Conclusion

Modern ships are increasingly equipped with a wide range of IT and OT components to enhance the safety and efficiency of marine operations. However, the growing reliance on electronic and computer-based systems also introduces significant cyber security risks. Today, ships are increasingly exposed to cyber threats, particularly GNSS attacks. To respond effectively to such attacks, both shore-based personnel and shipboard crews should receive comprehensive training. Moreover, their competence should be periodically evaluated.

The primary objective of this study is to design a comprehensive training programme that enables ship crews and shore-based personnel of ship operators to respond effectively to GNSS spoofing incidents and allows their responses to be observed during the drills. To achieve this objective, two separate ship-shore combined drills based on a GNSS spoofing scenario were conducted using the MarCy training programme, which is based on the CEM framework. Following the drills, the learners attended dedicated lecture sessions designed to reinforce the lessons learned. The effectiveness of the training programmes was evaluated through pre- and post-training knowledge assessments, enabling a quantitative comparison of learners' knowledge levels before and after the intervention.

A total of 20 learners from both shipboard and shore-based operational roles participated in two distinct training sessions. During the drills, interactions were established with 10 different third-party stakeholders, such as classification societies, marine insurers, and service providers. This approach also facilitated the assessment of the support capabilities of third-party stakeholders during a real GNSS spoofing incident. In addition, communication and coordination between shore-based personnel and third-party stakeholders were evaluated.

The study also presents findings and observations regarding the cyber security-related procedures and preparedness of the two participating ship operators. Finally, methodological contributions are provided through an assessment of the effectiveness of the MarCy training programme, together with recommendations for its further improvement and broader application.

The findings indicate that, despite increasing awareness of cyber security risks and the growing prevalence of GNSS-related incidents, both shipboard crews and shore-based personnel remain insufficiently prepared to respond effectively to GNSS attacks. In particular, significant challenges were observed in the practical implementation of the operators' existing procedures during the drills. Furthermore, the results revealed that third-party stakeholders may also be unable to provide adequate support when such attacks occur. Of the ten external stakeholders involved during the drills, only one was able to provide effective and actionable guidance in response to the simulated GNSS spoofing scenario.

Our review demonstrated that the operators' materials were predominantly IT-focused. There were no effective plans addressing the cyber risks associated with OT components, such as GNSS spoofing. Both shipboard and shore-based personnel had also been trained primarily on risks related to IT components. Training on the cyber risks associated with OT components was lacking. Although CySOs had been appointed in both shipboard and shore-based environments, it was observed that these individuals had not received any additional training beyond that provided to other personnel. It was further noted that existing documents and actual operator practices were not always aligned.

Finally, the effectiveness of the MarCy training programme was evaluated in this study. Learners' feedback and the findings obtained during the study confirm the validity of the recommendations incorporated into the MarCy training programme. All learners agreed that the training significantly increased their awareness of the cyber risks associated with GNSS technology. The MarCy training programme had previously been validated through contributions from domain experts and was subsequently evaluated and improved through learner participation. As a result of the present study, no major revisions to the MarCy training programme were identified as necessary. This outcome demonstrates that, while evaluation by domain experts is valuable in scientific studies, it is not sufficient on its own and should be complemented by the assessment of real-world operational outcomes.

During our study, the most significant challenge was finding ship operators willing to collaborate in conducting a ship-shore combined drill. Many ship operators did not respond to our invitation. Some kindly declined because they were concerned about potential reputational damage if the drill was unsuccessful, particularly in the presence of third-party stakeholders expected to participate.

At the beginning of this study, no specific attack type had been identified. As both ship operators collaborating in the cyber security drills preferred a GNSS spoofing scenario, the scope of the study was limited to GNSS spoofing attacks. Different cyber incident scenarios may lead to different findings. According to the operators' representatives, GNSS attacks are the cyber incidents with which both office personnel and ship crews are most familiar. Nevertheless, both shipboard and shore-based personnel experienced difficulties in responding effectively. Cyber incidents targeting other OT assets onboard may be even more challenging to handle.

Our study is limited to two cyber security drills conducted with the

participation of tanker operators. Therefore, the findings can not be generalised to ship operators managing other types of vessels, such as container ships, dry cargo vessels, and passenger ships. Moreover, the tanker operators involved in this study are subject to vetting programmes such as SIRE 2.0, CDI, and TMSA. These programmes are generally associated with higher operational and management standards. Therefore, the findings may differ for other ship operators.

Another limitation concerns the knowledge assessments used before and after the training. Although all six questions from the Quiz were repeated in the post-training Test, the Test also included four additional questions. Therefore, the two assessments were not fully equivalent. Moreover, their validity, reliability, difficulty, and other psychometric properties were not formally established. The observed increase in scores should therefore be interpreted as an indication of a change in learners' knowledge rather than as a standalone measure of training effectiveness. Nevertheless, the effectiveness of the training was not assessed solely based on the assessment scores. Learners' feedback was also collected through rating-scale, yes/no, and open-ended questions in the assessment survey. Considering these different sources of feedback together with the assessment results, the training may be considered effective.

A total of 20 learners participated in the training, seven of whom were shipboard crew members. In addition, 10 different stakeholders were involved in the study. Therefore, training sessions conducted with different learners and different stakeholders may yield different findings, observations, and recommendations.

Based on the observations and findings of the study, several recommendations are proposed for maritime stakeholders. Regulatory authorities should mandate regular cyber security drills. Currently, IMO regulations do not include cyber security drills. However, local maritime authorities may require such drills for ship operators. For instance, the USCG requires a minimum of two cyber security drills per year for U.S.-flagged ships. Similar to traditional emergency drills, the establishment of international requirements for cyber incident drills would help ensure a minimum level of preparedness across the maritime sector.

Ship operators should conduct comprehensive gap analyses of their SMS, SSP, and SSA. Procedures and checklists should address threats targeting OT systems, such as GNSS spoofing. Clear guidance on detection, reporting, and response actions should be provided. In addition, ship operators should increase both the frequency and effectiveness of their cyber security training programmes. Rather than relying on infrequent and generic training, frequent role-based training sessions should be implemented. In particular, personnel responsible for cyber security should receive advanced training that goes beyond basic awareness-level instruction.

Finally, strong collaboration among maritime stakeholders, such as service providers, marine insurers, maritime authorities, and classification societies, is recommended. Establishing joint drills or information-sharing agreements with stakeholders would enhance preparedness for cyber incidents.

In future studies, drills could be conducted using different cyber attack scenarios. However, these scenarios should focus on OT-specific attacks to address the needs of the maritime industry. Both ship operators stated that they require a cyber security gap analysis covering their SMS, SSP, SSA, and marine insurance policies. In future studies, conducting such a gap analysis would allow company materials to be examined in greater detail, as well as enabling an evaluation of the effectiveness of these procedures in practice. Moreover, although the training strategies recommended by the MarCy programme, such as drills, lectures, discussions, and case studies, have been shown to be effective in the comprehensive training sessions conducted, the Demonstration approach has not yet been evaluated. Future studies incorporating the Demonstration approach may further enhance the design of maritime cyber security training and enable an assessment of its effectiveness. Future studies could also focus on developing measurable competency and success criteria for the MarCy training

programme. Such criteria could be evaluated and refined through training sessions involving different ship operators, learner groups, and cyber incident scenarios.

CRedit authorship contribution statement

Aybars Oruc: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

References

- Aris, S.R.S., Isa, W.A.R.W.M., Yahaya, W.A.W., Mohamad, S.N.A., 2017. Multidisciplinary curriculum design approaches towards balanced and holistic graduates. In: 2017 IEEE 9th International Conference on Engineering Education (ICEED). Kanazawa, Japan. IEEE, 09 November 2017 - 10 November 2017.
- BIMCO, 2021. ClassNK, CSA, DCSA, ICS, INTERCARGO, INTERMANAGER, INTERTANKO, IMCA, IUMI, e t al. The Guidelines on Cyber Security Onboard Ships. <https://www.ics-shipping.org/resource/guidelines-on-cyber-security-onboard-ships-version-four/>. accessed on 5 July 2026.
- BIMCO, 2024. ClassNK, CSA, DCSA, ICS, INTERCARGO, INTERMANAGER, INTERTANKO, IMCA, IUMI, e t al. The guidelines on cyber security onboard ships. <https://www.ocimf.org/publications/information-papers/the-guidelines-on-cyber-security-onboard-ships>. accessed on 5 July 2026.
- Bobbitt, F., 2009. Scientific method in curriculum-making. In: Flinders, D.J., Thornton, S. J. (Eds.), *The Curriculum Studies Reader*, 3rd ed. Routledge, New York, USA, pp. 15–21.
- Braun, V., Clarke, V., 2006. Using thematic analysis in psychology. *Qual. Res. Psychol.* 3, 77–101. <https://doi.org/10.1191/1478088706qp0630a>.
- BV, 2024. NR659 R04 Rules on cyber security for the classification of marine units. BV: Courbevoie, France. <https://rules.veristar.com/dy/data/bv/pdf/659-NR-2024-07.pdf>. accessed on 5 July 2026.
- C4ADS, 2019. Above us only stars. Available online. <https://c4ads.org/wp-content/uploads/2022/05/AboveUsOnlyStars-Report.pdf>. accessed on 5 July 2026.
- CDI, 2023. Ship Inspection Report: 10th Edition SIR Core Questionnaire, 10.5th ed. Available online. https://www.cdim.org/webshop/cdi/fr_cdim.jsp accessed on 5 July 2026.
- 33 CFR Part 101 Subpart F Cybersecurity, 2025.
- ClassNK, 2024. Guidelines for cyber resilience of ships. ClassNK: Tokyo, Japan. https://www.classnk.or.jp/hp/pdf/activities/cybersecurity/gl_CyberResilienceofShips_202407e.pdf. accessed on 5 July 2026.
- Coach RJ. Ship to Shore drill onboard training. Available Online, <https://www.youtube.com/watch?v=6qR2Rnan25o>, 2021, accessed on 5 July 2026.
- Cybersecurity in the Marine Transportation System, 2025.
- Cydome, 2026. Tankers Collide in the Gulf, Analysts Suspect GPS Jamming. Available online. <https://cydome.io/tankers-collide-in-the-gulf-analysts-suspect-gps-jamming/>. accessed on 5 July 2026.
- Dalkley, N., Helmer, O., 1963. An experimental application of the Delphi method to the use of experts. *Manag. Sci.* 9, 458–467.
- Deng, Z.L., Liu, H.D., Huang, J.M., Zou, D.J., 2009. A novel anti-jam navigation system based on A-GNSS. In: 2009 Joint Conferences on Pervasive Computing (JPCP). Tamsui, Taipei, Taiwan, 2009. IEEE, pp. 279–284, 03–05 Dec. ISBN 978-1-4244-5227-9.
- Denizcilik, G., 2026. Ship-Shore combined ISPS drill. Available online. <https://www.youtube.com/watch?v=YfabWVZomiM>. accessed on 5 July.
- Diakun, B., 2026. Russia's far east ports go dark in AIS interference outbreak. Available online: <https://www.lloydslist.com/LL1154502/Russia%20%80%99s-far-east-ports-go-dark-in-AIS-interference-outbreak>. accessed on 5 July 2026.
- DNV, 2024. DNV-CG-0325 Cyber secure. DNV: Høvik, Norway. <https://standards.dnv.com/explorer/>. accessed on 5 July 2026.
- DSCA, 2020. Implement. guide cyber secur. Vessels. Available online. <https://dcsa.org/wp-content/uploads/2020/03/DCSA-Implementation-Guideline-for-BIMCO-Compliant-Cyber-Security-on-Vessels-v1.0.pdf>. accessed on 5 July 2026.
- Erasmus, B., van Dyk, P., 1999. *Training Management in South Africa*, 2nd ed. Oxford University Press, Cape Town, South Africa. ISBN 0 19 571897 6.
- Erstad, E., Hopcraft, R., Vineetha Harish, A., Tam, K., 2023. A human-centred design approach for the development and conducting of maritime cyber resilience training. *WMU J. Marit. Aff.* 22, 241–266. <https://doi.org/10.1007/s13437-023-00304-7>.
- Filic, M., 2018. Foundations of GNSS spoofing detection and mitigation with distributed GNSS SDR receiver. *TransNav Int. J. Mar. Navig. Saf. Sea Transp.* 12, 649–656. <https://doi.org/10.12716/1001.12.04.01>.
- French, S., 2015. The benefits and challenges of modular higher education curricula. *Issues and ideas paper. Melb. Melb. Cent. Study High. Educ.* 1–12.
- Gahnström, J., 2026. Free article: Jamming and Spoofing. Available online. <https://www.nautinst.org/resources-page/free-article-jamming-and-spoofing.html>. accessed on 5 July 2026.
- Glomsvoll, O., Bonenberg, L.K., 2017. GNSS jamming resilience for close to shore navigation in the Northern Sea. *J. Navig.* 70, 33–48. <https://doi.org/10.1017/S0373463316000473>.
- Goward, D., 2026. Mass GPS spoofing attack in black sea? Available online. <https://www.maritime-executive.com/editorials/mass-gps-spoofing-attack-in-black-sea> (accessed on 5 July 2026).
- GPSwise, 2026. Live GPS spoofing & jamming map. Available online. <https://gpswise.aero/>. accessed on 5 July 2026.
- Graham, L., 2026. Shipping industry vulnerable to cyber attacks and gps jamming. Available online. <https://www.cnn.com/2017/02/01/shipping-industry-vulnerable-to-cyber-attacks-and-gps-jamming.html>. accessed on 5 July 2026.
- Grammarly, 2026. Our features. Available online. <https://www.grammarly.com/features>. accessed on 5 July 2026.
- Höller, L., 2026. Researchers home in on origins of Russia's baltic GPS jamming. Available online. <https://www.defensenews.com/global/europe/2025/07/02/researchers-home-in-on-origins-of-russias-baltic-gps-jamming/>. accessed on 5 July 2026.
- IACS, 2023. E26 Cyber Resilience of Ships, 1st ed. IACS, London, UK. Available online. <https://iacs.org.uk/resolutions/unified-requirements/ur-e>. accessed on 5 July 2026.
- IACS, 2026. About us. Available online. <https://iacs.org.uk/about-us/>. accessed on 5 July 2026.
- Ibanez, D., Rovira-Garcia, A., Sanz, J., Juan, J.M., Gonzalez-Casado, G., Jimenez-Banos, D., Lopez-Echazarreta, C., 2018. Lapin, I. the GNSS laboratory tool suite (gLAB) updates: SBAS, DGNSS and global monitoring system. In: 2018 9th ESA Workshop on Satellite Navigation Technologies and European Workshop On GNSS Signals and Signal Processing (NAVITEC), 2018. IEEE, Noordwijk, Netherlands, pp. 1–11, 05–07 Dec. ISBN 978-1-5386-7125-2.
- IMO, 2000. Resolution MSC.112(73) Adoption of the Revised Performance Standards For Shipborne Global Positioning System (GPS) Receiver Equipment. IMO, London, UK.
- IMO, 2002a. ISPS Code: Part A: 13 Training, Drills and Exercises On Ship Security. IMO, London, UK.
- IMO, 2002b. ISPS Code: Part B: 13 Training, Drills and Exercises On Ship Security. IMO, London, UK.
- IMO, 2014. Resolution A.1079(28) Recommendations for the Training and Certification of Personnel on Mobile Offshore Units (MOUs). IMO, London, UK.
- IMO, 2017a. SOLAS chapter III life saving appliances and arrangements: regulation 19 emergency training and drills. IMO, London, UK.
- IMO, 2017b. Resolution MSC.428(98) Maritime Cyber Risk Management In Safety Management Systems. IMO, London, UK.
- IMO, 2021. Guide to Maritime Security and the ISPS Code: Section 4 Security Responsibilities of Ship Operators - 4.13 Cyber Security on Board Ships. IMO, London, UK.
- IMO, 2025. MSC-FAL.1/Circ.3/Rev.3 Guidelines on Maritime Cyber Risk Management. IMO, London, UK.
- de La Vallée, P., Iosifidis, G., Rossi, A., Dri, M., Mees, W., 2022. Sector-specific training - a federated maritime scenario. In: Dziech, A., Mees, W., Niemiec, M. (Eds.), *Multimedia Communications, Services and Security*. Springer International Publishing, Cham, pp. 21–35. ISBN 978-3-031-20214-8.
- Jan, S.-S., Tao, A.-L., 2016. Comprehensive comparisons of satellite data, signals, and measurements between the BeiDou navigation satellite system and the global positioning system. *Sensors* 16, 689. <https://doi.org/10.3390/s16050689>.
- Liu, Y., Li, S., Fu, Q., Liu, Z., 2018. Impact assessment of GNSS spoofing attacks on INS/GNSS integrated navigation system. *Sens. (Basel)* 18, 1433. <https://doi.org/10.3390/s18051433>.
- Lovell, K.N., Heering, D., 2019. Exercise Neptune: maritime cybersecurity training using the navigational simulators. In: 5th Interdisciplinary Cyber Research Conference. Tallinn, Estonia. Jun. 29, 2019.
- M-CERT, 2026. ADMIRAL dataset. Available online. <https://www.m-cert.fr/admiral/>. accessed on 5 July 2026.
- Mirriahi, N., Alonzo, D., Fox, B., 2015. A blended learning framework for curriculum design and professional development. *Res. Learn. Technol.* 23, 23. <https://doi.org/10.3402/rlt.v23.28451>.
- Nadler, L., Nadler, Z., 1982. *Designing Training programs: The Critical Events Model*, 1st ed. Addison-Wesley Publishing, US.
- NATO Shipping Centre, 2026. <https://shipping.nato.int/nsc/page10303037.aspx> accessed on 5 July 2026.
- NHL Stenden, 2026. <https://maritim cybersecurity.nl/> (accessed on 5 July 2026).
- OCIMF. Tmsa, Ocimf: Scotland, UK, 2017.

- OCIMF, 2022. SIRE 2.0 Question library: Part 1 - Chapters 1 to 7, 1st ed. OCIMF, London, UK. Available online. <https://www.ocimf.org/document-library/630-sire-2-0-question-library-part-1-chapters-1-to-7-version-1-0-january-2022/file>. accessed on 5 July 2026.
- Oruc, A., 2020. Claims of state-sponsored cyberattack in the maritime industry. In: *The International Naval Engineering Conference and Exhibition (INEC 2020)*, 05–09 Oct. 2020.
- Oruc, A., Chowdhury, N., Gkioulos, V., 2024a. A modular cyber security training programme for the maritime domain. *Int. J. Inf. Secur.* 23, 1477–1512. <https://doi.org/10.1007/s10207-023-00799-4>.
- Oruc, A., Chowdhury, N., Gkioulos, V., Katsikas, S., 2024b. Evaluation of maritime cyber security (MarCy) training programme. *TransNav Int. J. Mar. Navig. Saf. Sea Transp.* 18, 743–763. <https://doi.org/10.12716/1001.18.04.01>.
- Oruc, A., Gkioulos, V., Katsikas, S., 2022. Towards a cyber-physical range for the integrated navigation system (INS). *J. Mar. Sci. Eng.* 10, 107. <https://doi.org/10.3390/jmse10010107>.
- Oruc, A., Kavallieratos, G., Gkioulos, V., Katsikas, S., 2025. Perspectives on the cybersecurity of the integrated navigation system. *J. Mar. Sci. Eng.* 13, 1087. <https://doi.org/10.3390/jmse13061087>.
- Potamos, G., Theodoulou, S., Stavrou, E., Stavrou, S., 2023. Building maritime cybersecurity capacity against ransomware attacks. In: Onwubiko, C., Rosati, P., Rege, A., Erola, A., Bellekens, X., Hindy, H., Jaatun, M.G. (Eds.), *Proceedings of the International Conference on Cybersecurity, Situational Awareness and Social Media*. Singapore. Springer Nature Singapore, pp. 87–101. ISBN 978-981-19-6413-8.
- ProWritingAid, 2026. Features. Available online. <https://prowritingaid.com/features>. accessed on 5 July 2026.
- QualCoder, 2026. What is QualCoder. Available online. <https://qualcoder.wordpress.com/reports/>. accessed on 5 July 2026.
- RightShip, 2025. RightShip inspection ship questionnaire (RISQ), 3.2nd ed. Available online. <https://rightship.com/solutions/shipowner/rightship-inspection-ship-questionnaire-risq>. accessed on 5 July 2026.
- Schagaev, I., Bacon, E., Ioannides, N., 2010. An approach to curriculum design for computer science. In: Ordonez De Pablos, P., Ziderman, A., Roulstone, A., Maurer, H., Imber, J.B. (Eds.), *Knowledge Management, Information Systems, E-Learning, and Sustainability Research*, Lytras, M.D. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 493–499. ISBN 978-3-642-16317-3.
- Scribd, 2026. Maritime cybersecurity guide. Available online. <https://www.scribd.com/document/714891472/CYBER-SECURITY-MANUAL-ENG>. accessed on 5 July 2026.
- Singh, S., Kumar, R.K., Sundaram, K.R., Kanjilal, B., Nair, P., 2012. Improving outcomes and reducing costs by modular training in infection control in a resource-limited setting. *Int. J. Qual. Health Care* 24, 641–648. <https://doi.org/10.1093/intqhc/mzs059>.
- SQE Marine, 2026. ISM ship-shore drills pack. Available online. <https://sqemarine.com/wp-content/uploads/Manuals/ISM-Drill-Pack/ISM%20Drills%20pack/assets/basic-html/page-1.html>. accessed on 5 July 2026.
- Svilicic, B., Rudan, I., Francić, V., Mohović, D., 2020. Towards a cyber secure shipboard radar. *J. Navig.* 73, 547–558. <https://doi.org/10.1017/S0373463319000808>.
- Świerczyński, S., Zwolan, P., Rutkowska, I., 2016. Jamming as a threat to navigation. *Annu. Navig.* 23, 219–233. <https://doi.org/10.1515/aon-2016-0016>.
- Tam, K., Moara-Nkwe, K., Jones, K., 2020. The use of cyber ranges in the maritime context. *Marit. Technol. Res.* 3. <https://doi.org/10.33175/mtr.2021.241410>.
- USCG Navigation Center, 2026. GPS problem report. Available online. <https://www.navcen.uscg.gov/contact/gps-problem-report>. accessed on 5 July 2026.
- Warner, J.S., Johnston, R.G., 2003. GPS spoofing countermeasures. *Homel. Secur. J.* 25, 19–27.
- Wiseman, J., 2026. GPSJAM. Available online. <https://gpsjam.org/> (accessed on 5 July 2026).
- Witherbys, BIMCO, ICS. cyber security workbook for on board ship use, 2025.